



ANALYSIS OF PRESSURE CHARACTERISTICS OF PROPELLER BLADE

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September, 2017

This thesis is submitted to the Department of Naval Architecture and Marine Engineering, Bangladesh University of Engineering and Technology, Dhaka, in partial fulfillment of the requirement for the degree of **Bachelor of Science in Naval Architecture and Marine Engineering.**

CANDIDATE'S DECLARATION

None of the works contained in this thesis has been submitted in support of an application for another degree or qualification of this or any other University or Institution.

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ACKNOWLEDGMENT

All praise and thanks is due to the Almighty. We would like to express our heartfelt gratitude to Dr. Md. Refayet Ullah, Professor, Department of Naval Architecture And Marine Engineering, BUET, Dhaka. Our thesis supervisor for his patient guidance, enthusiastic encouragement and useful critiques of this thesis work. His all constructive evaluation of our work helped us completing the thesis on time. We would also like to thank Dr. Gautam Kumar Saha and Dr. Md. Shahidul Islam for their advice and assistance in keeping our progress on schedule. Without the co-ordination of the partners this would not have been possible.

Last but not the least, we wish to thank our parents for their support and encouragement throughout our study.

ABSTRACT

Propeller efficiency mostly depends on the pressure distribution around the blade sections. This thesis incorporates some basic ideas about the pressure characteristics to run some calculations to figure out how pressure is distributed and plot several graphs for different blade sections at different radius from hub to the tip of a propeller blade. Using some basic analytical procedures we ended up getting the pressure distribution along with K_T , K_Q and resulting propeller efficiency. The values obtained through calculations are validated by comparing it with that of Burrill analysis. These calculations are then followed by the procedure to figure out if cavitation is taking place.

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CHAPTER 1

INTRODUCTION

In nowadays, to achieve the best propulsion performance for commercial ship owner, an excellent propeller design is one of the most economical saving ways. A large numbers of factors may affect the characteristics and performance of propellers e.g. the levels of periodic forces, blade structural strengths, cavitation, as well as the noises and vibrations induced by the propellers and last but not the least is the pressure distribution around the blades of a propeller. Among all these effects pressure distribution and cavitation check would be investigated in this thesis.

There are many softwares to perform the very tasks contained in this thesis ,but the sole purpose of this thesis is to go in an analytical way which can be compared with the obtained data from software simulation .

1.1 OBJECTIVE

- Generating pressure distribution around a propeller blade sections and plotting respective graphs.
- Checking if cavitation is taking place on the blade section.

1.2 METHODOLOGY

First of a model propeller 4118 has been chosen to carry out the entire thesis and corresponding propeller geometry was provided within the catalog. With the help of it all the necessary calculations were done to generate pressure distribution which is the sole purpose of this thesis.

The flow-chart showing the steps we took to get to the very end of this thesis is in the following

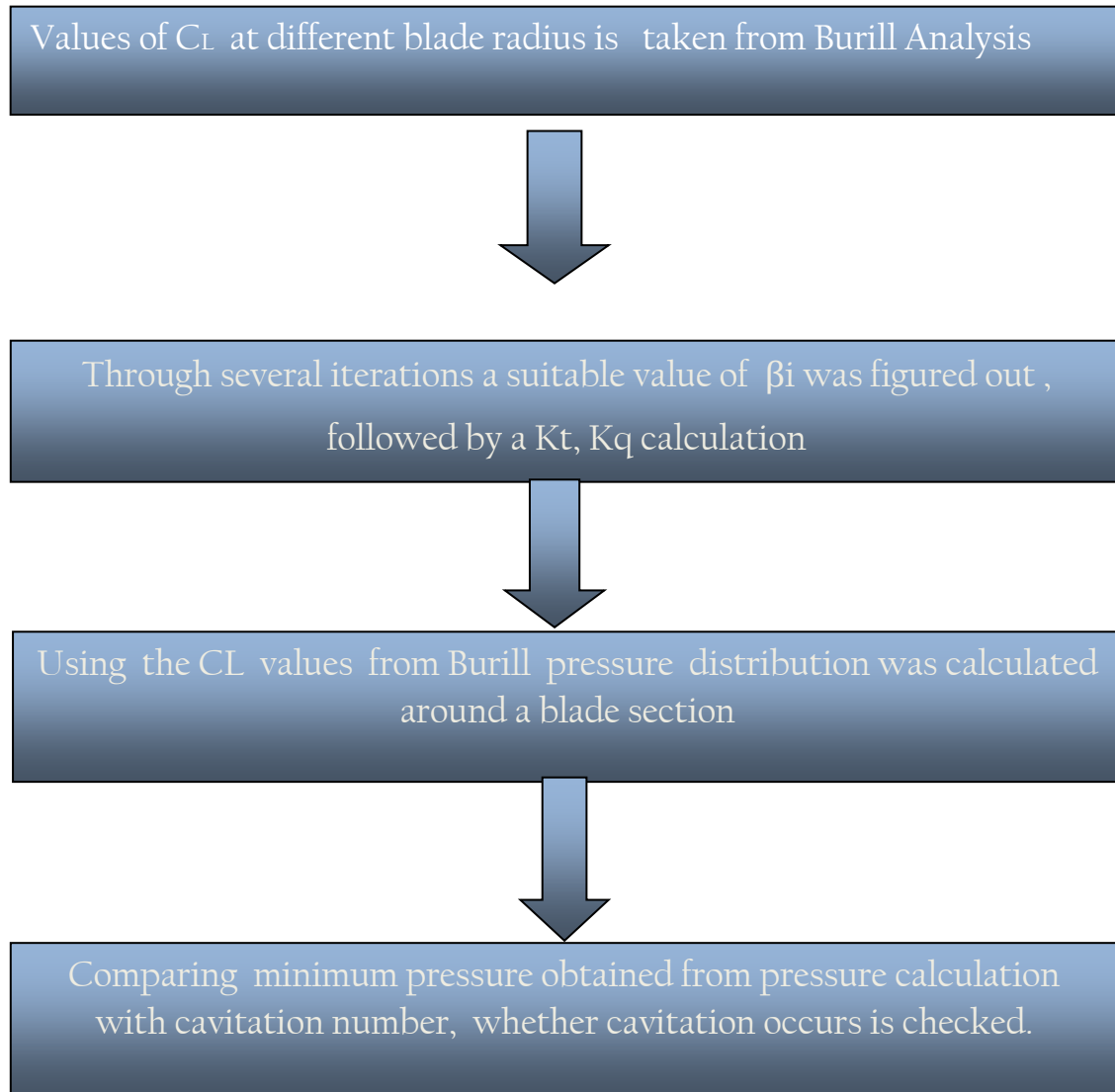


Figure 1.1: Flow Chart

CHAPTER 2

2.1 Existing data

The followings are the main data of a model 4118 propeller

Diameter $D=304.8\text{mm}=0.3048\text{m}$

Pitch ratio at $0.7R=1.077$

Number of blades=3

Expanded area ratio=.606

Mean width ratio=.397

Blade thickness friction= 0.04

NACA Nu line, $a=0.8$

NACA Section shape=66(mod)

Speed of advance, $V_A=1\text{ m/s}$

Revolution per second, $n=3.94\text{ r.p.s}$

2.2 Propeller model 4118 Geometry

X (Along chord length)	Chord Ratio C/D	Thickness Ratio t/D	Pitch Ratio P/D	Camber Ratio m/C
0.2	0.32	0.0439	1.086	0.0224
0.3	0.3636	0.0282	1.084	0.0232
0.4	0.4048	0.0239	1.082	0.0233
0.5	0.4392	0.0196	1.08	0.0218
0.6	0.461	0.016	1.078	0.0207
0.7	0.4622	0.0125	1.077	0.02
0.8	0.4347	0.0091	1.075	0.0197
0.9	0.3613	0.006	1.073	0.0182
0.95	0.2775	0.0045	1.07	0.0167

2.3 Open water Characteristics

Advance co-efficient J	Thrust co-efficient K_T	Torque co-efficient K_Q	Efficiency η
0.4	0.3575	0.055	0.4143
0.5	0.3144	0.0502	0.4986
0.6	0.2695	0.0448	0.5747
0.7	0.222	0.0386	0.6415
0.8	0.1713	0.0314	0.6957
0.9	0.1181	0.0233	0.7263
1	0.0629	0.0144	0.6931

2.4 Values from Burrill Analysis

J	x	C _L	C _D
0.7 (Part Load)	0.3	0.3485	0.01488
	0.5	0.2972	0.01255
	0.7	0.2421	0.01106
	0.9	0.1798	0.0097
0.833 (Full Load)	0.3	0.2606	0.01314
	0.5	0.2247	0.00932
	0.7	0.1865	0.01017
	0.9	0.1482	0.00953
1 (Over Load)	0.3	0.1406	0.01144
	0.5	0.1311	0.01085
	0.7	0.1204	0.01006
	0.9	0.1061	0.00986

2.5 Advance co-efficient Vs Efficiency & NACA section drawing

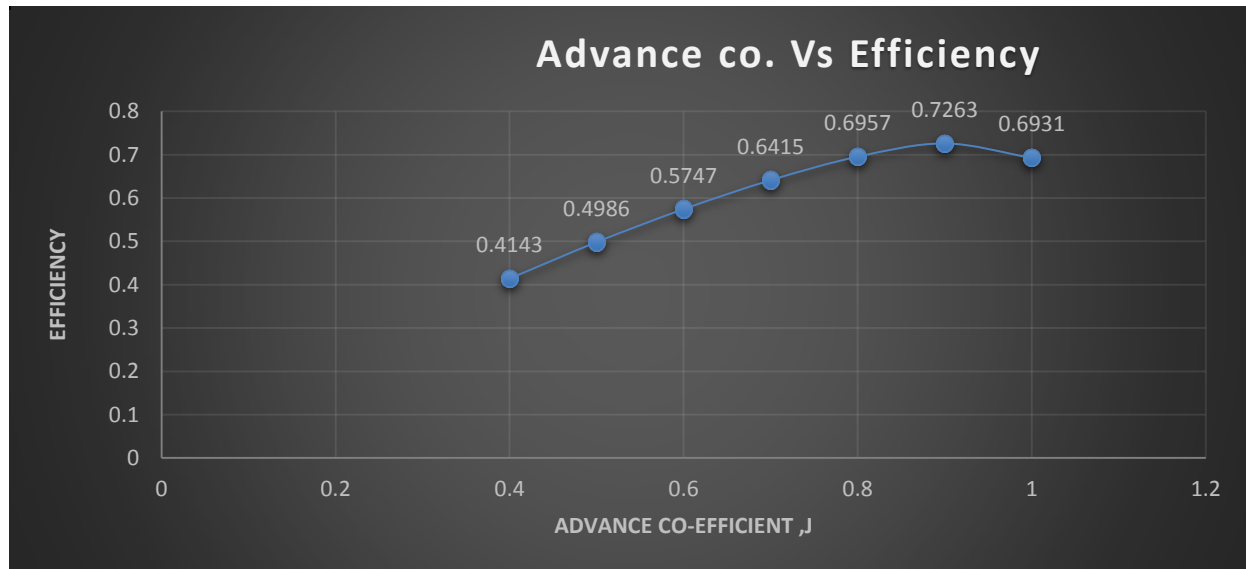


Figure 2.1: Advance co-efficient Vs Efficiency

Advance Co-efficient, $J = V_A / n D = 0.833$

Efficiency, $\eta = 0.705798$ (from graph)

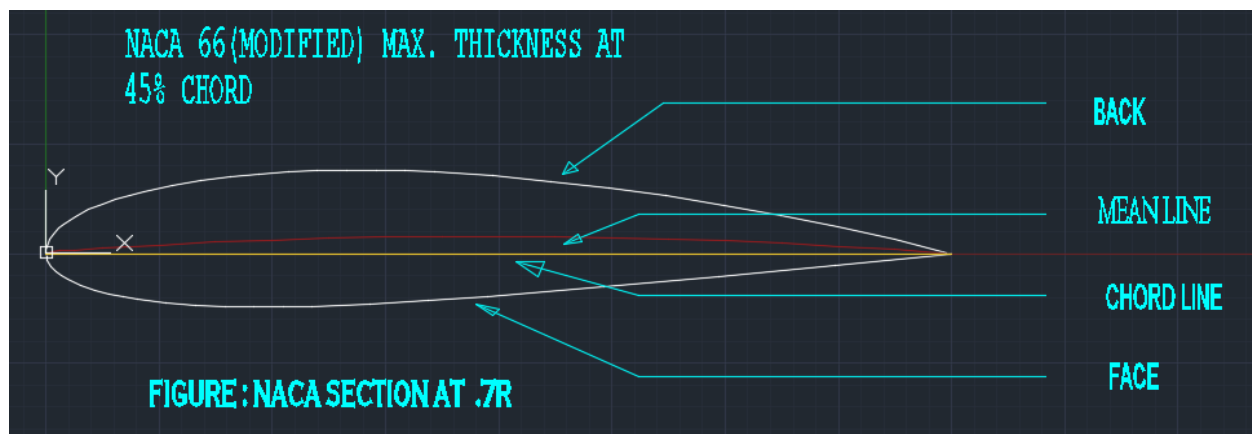


Figure 2.2: NACA 66 (modified) on autocad

CHAPTER 3

THRUST, TORQUE & EFFICIENCY CALCULATION

3.1 Expressions to determine propeller performance parameters

Goldstein Correction Factor, $k = (2/\pi) \cos^{-1} [e^{((z/2) \cdot \frac{1-x}{x \sin \beta_i})}] \dots \dots \dots (1)$

The section Lift Co-efficient, $C_L = (4\pi/z)(D/C) \cdot x \cdot k \cdot \sin \beta_i \cdot \tan(\beta_i - \beta) \dots \dots \dots (2)$

Combining these two equations we get

$$C_L = (4\pi/z)(D/C) \cdot x \cdot (2/\pi) \cos^{-1} [e^{((z/2) \cdot \frac{1-x}{x \sin \beta_i})}] \cdot \sin \beta_i \cdot \tan(\beta_i - \beta)$$

$$\text{Or, } C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1} \{e^{((z/2) \cdot \frac{1-x}{x \sin \beta_i})}\}] \cdot \sin \beta_i \cdot \tan(\beta_i - \beta) = 0 \dots \dots \dots (4)$$

The thrust and torque produced by an element for all the z-blades are dT and dQ, then from figure 3.1,

Thrust grading, $dT/dr = z \cdot C_L \cdot .5 \cdot \rho \cdot C \cdot V_R^2 \cdot \cos \beta_i (1 - \tan \beta_i \cdot \tan \gamma) \dots \dots \dots (5)$

Torque grading, $dQ/dr = r \cdot z \cdot C_L \cdot .5 \cdot \rho \cdot C \cdot V_R^2 \cdot \cos \beta_i (\tan \beta_i + \tan \gamma) \dots \dots \dots (6)$

From equation (5) we get,

$$\begin{aligned} \frac{dKT}{dx} &= \frac{dT}{dr} \left(\frac{dr}{dx} \cdot \frac{1}{\rho n^2 D^4} \right) \\ &= \frac{dT}{dr} \cdot \frac{R}{\rho n^2 D^4} \\ &= \frac{dT}{dr} \cdot \frac{1}{2\rho n^2 D^3} \\ &= z \cdot C_L \cdot .5 \cdot \rho \cdot C \cdot V_R^2 \cdot \cos \beta_i (1 - \tan \beta_i \cdot \tan \gamma) \cdot \frac{1}{2\rho n^2 D^3} \\ &= z \cdot C_L \cdot \frac{C/D}{4n^2 D^2} \cdot V_R^2 \cdot \cos \beta_i (1 - \tan \beta_i \cdot \tan \gamma) \end{aligned}$$

Similarly from equation (6) we get,

$$\frac{dKQ}{dx} = r \cdot z \cdot C_L \cdot \frac{C/D}{4n^2 D^3} \cdot V_R^2 \cdot \cos \beta_i (\tan \beta_i + \tan \gamma)$$

Where,

$$V_R = V_A \cdot \frac{\cos(\beta_i - \beta)}{\sin(\beta)} \text{ and } \tan \gamma = C_D/C_L$$

3.2 Calculation of β_i , V_R , $\frac{dKT}{dx}$, $\frac{dKQ}{dx}$

3.2.1 For Full load ($J=0.833$):

$$x=.3$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A=1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n = V_A/JD = \frac{1}{.833*0.3048} = 3.94$$

$$\tan\beta = V_A/(n\pi D x) = \frac{1}{3.94*3.1416*0.3048*.3} = .8835$$

$$\beta = 41.46 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2)\frac{1-x}{x\sin\beta_i})}\}] * \sin\beta_i \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .2606 - \frac{8*.3}{3*3.636*57.2958} [\cos^{-1}\{e^{((-3/2)(\frac{1-.3}{.3*\sin\beta_i})}\}] * \sin\beta_i \tan(\beta_i - 41.46) = 0$$

Solving for β_i , $\beta_i = 47.34 \text{ deg.}$

Calculation for V_R :

$$\begin{aligned} V_R &= V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)} \\ &= 1 * \frac{\cos(47.34 - 41.46)}{\sin(41.46)} \\ &= 1.5024 \text{ m/s} \end{aligned}$$

Calculation for $\frac{dKT}{dx}$:

$$\begin{aligned} \frac{dKT}{dx} &= z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos\beta_i (1 - \tan\beta_i * \tan\gamma) \\ \frac{dKT}{dx} &= 3 * .2606 * \frac{.3636}{4*3.94^2*0.3048^2} * 1.5024^2 * \cos(47.34) (1 - \tan(47.34) * \tan(.01314/.2606)) \\ &= .0753 \end{aligned}$$

Calculation for $\frac{dKQ}{dx}$:

$$\begin{aligned} \frac{dKQ}{dx} &= r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos\beta_i * (\tan\beta_i + \tan\gamma) \\ &= (.3048*3/2) * 3 * .2606 * \frac{.3636}{4*3.94^2*0.3048^3} * 1.5024^2 * \cos(47.34) (\tan(47.34) + \tan(.01314/.2606)) \\ &= .1227 \end{aligned}$$

$$x=.5$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A = 1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n = V_A/JD = \frac{1}{.833*.3048} = 3.94$$

$$\tan\beta = V_A/(n\pi D x) = \frac{1}{3.94*3.1416*.3048*.5} = .5301$$

$$\beta = 27.93 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2)(\frac{1-x}{x\sin\beta_i})}\}]\sin\beta_i \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .2247 - \frac{8*.5}{3*.4392*57.2958} [\cos^{-1}\{e^{((-3/2)(\frac{1-.5}{.5*\sin\beta_i})}\}]\sin\beta_i \tan(\beta_i - 27.93) = 0$$

Solving for β_i ,

$$\beta_i = 33.07 \text{ deg.}$$

Calculation for V_R :

$$V_R = V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)}$$

$$= 1 * \frac{\cos(33.07 - 27.93)}{\sin(27.93)}$$

$$= 2.1264 \text{ m/s}$$

Calculation for $\frac{dKT}{dx}$:

$$\frac{dKT}{dx} = z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos\beta_i (1 - \tan\beta_i * \tan\gamma)$$

$$\frac{dKT}{dx} = 3 * .2247 * \frac{.4392}{4*3.94^2*.3048^2} * 2.1264^2 * \cos(33.07) (1 - \tan(33.07) * \tan(.00932/.2247))$$

$$= .19437$$

Calculation for $\frac{dKQ}{dx}$:

$$\frac{dKQ}{dx} = r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos\beta_i * (\tan\beta_i + \tan\gamma)$$

$$= (.3048*.5/2) * 3 * .2247 * \frac{.4392}{4*3.94^2*.3048^3} * 2.1264^2 * \cos(33.07) (\tan(33.07) + \tan(.00932/.2247))$$

$$= .0317$$

$$x=.7$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A = 1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n = V_A/JD = \frac{1}{.833*.3048} = 3.94$$

$$\tan\beta = V_A/(n\pi D x) = \frac{1}{3.94*3.1416*.3048*.7} = .3787$$

$$\beta = 20.74 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2)\frac{1-x}{x\sin\beta_i})}\}] * \sin\beta_i \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .1865 - \frac{8*.7}{3*.4622*57.2958} [\cos^{-1}\{e^{((-3/2)\frac{1-.7}{.7*\sin\beta_i})}\}] * \sin\beta_i \tan(\beta_i - 20.74) = 0$$

Solving for β_i ,

$$\beta_i = 25.208 \text{ deg.}$$

Calculation for V_R :

$$V_R = V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)}$$

$$= 1 * \frac{\cos(25.208 - 20.74)}{\sin(20.74)}$$

$$= 2.8153 \text{ m/s}$$

Calculation for $\frac{dKT}{dx}$:

$$\frac{dKT}{dx} = z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos\beta_i (1 - \tan\beta_i * \tan\gamma)$$

$$\frac{dKT}{dx} = 3 * .1865 * \frac{.4622}{4*3.94^2*.3048^2} * 2.8153^2 * \cos(25.208) (1 - \tan(25.208) * \tan(0.01017/0.1865))$$

$$= .3132$$

Calculation for $\frac{dKQ}{dx}$

$$\frac{dKQ}{dx} = r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos\beta_i * (\tan\beta_i + \tan\gamma)$$

$$= (.3048*7/2) * 3 * .1865 * \frac{.4622}{4*3.94^2*.3048^3} * 2.8153^2 * \cos(25.208) (\tan(25.208) + \tan(0.01017/0.1865))$$

$$= .0591$$

$$x=.9$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A = 1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n = V_A/JD = \frac{1}{.833*0.3048} = 3.94$$

$$\tan\beta = V_A/(n\pi D x) = \frac{1}{3.94*3.1416*0.3048*.9} = .2945$$

$$\beta = 16.41 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2)\frac{1-x}{x\sin\beta_i})}\}] * \sin\beta_i \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .1482 - \frac{8*.9}{3*3.613*57.2958} [\cos^{-1}\{e^{((-3/2)\frac{1-.9}{.9*\sin\beta_i})}\}] * \sin\beta_i \tan(\beta_i - 16.41) = 0$$

Solving for β_i ,

$$\beta_i = 20.46 \text{ deg.}$$

Calculation for V_R :

$$\begin{aligned} V_R &= V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)} \\ &= 1 * \frac{\cos(20.46 - 16.41)}{\sin(16.41)} \\ &= 3.5309 \text{ m/s} \end{aligned}$$

Calculation for $\frac{dKT}{dx}$:

$$\begin{aligned} \frac{dKT}{dx} &= z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos\beta_i (1 - \tan\beta_i * \tan\gamma) \\ \frac{dKT}{dx} &= 3 * .1482 * \frac{.3613}{4*3.94^2*0.3048^2} * 3.5309^2 * \cos(20.46) (1 - \tan(20.46) * \tan(.00953/.1482)) \\ &= .3252 \end{aligned}$$

Calculation for $\frac{dKQ}{dx}$:

$$\begin{aligned} \frac{dKQ}{dx} &= r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos\beta_i * (\tan\beta_i + \tan\gamma) \\ &= (.3048*.9/2) * 3 * .1482 * \frac{.3613}{4*3.94^2*0.3048^3} * 3.5309^2 * \cos(20.46) (\tan(20.46) + \tan(.00953/.1482)) \\ &= .0548 \end{aligned}$$

3.2.1.1 Table for K_T, K_Q, η for $J=0.833$

x	dK _T /dx	SM	Product	dK _Q /dx	SM	Product
0.2	0.02	1	0.02	0.004	1	0.004
0.3	0.0753	4	0.3012	0.01227	4	0.04908
0.4	0.13	2	0.26	0.0226	2	0.0452
0.5	0.1944	4	0.7776	0.0317	4	0.1268
0.6	0.26	2	0.52	0.0452	2	0.0904
0.7	0.3132	4	1.2528	0.0591	4	0.2364
0.8	0.336	2	0.672	0.0607	2	0.1214
0.9	0.3252	4	1.3008	0.0548	4	0.2192
1	0	1	0	0	1	0
			5.1044			0.89248

$$K_T = 0.170147$$

$$K_Q = 0.029749$$

$$\eta = 0.758246$$

3.2.1.2 : Graph for x Vs dK_T/dx , dK_Q/dx , β_i

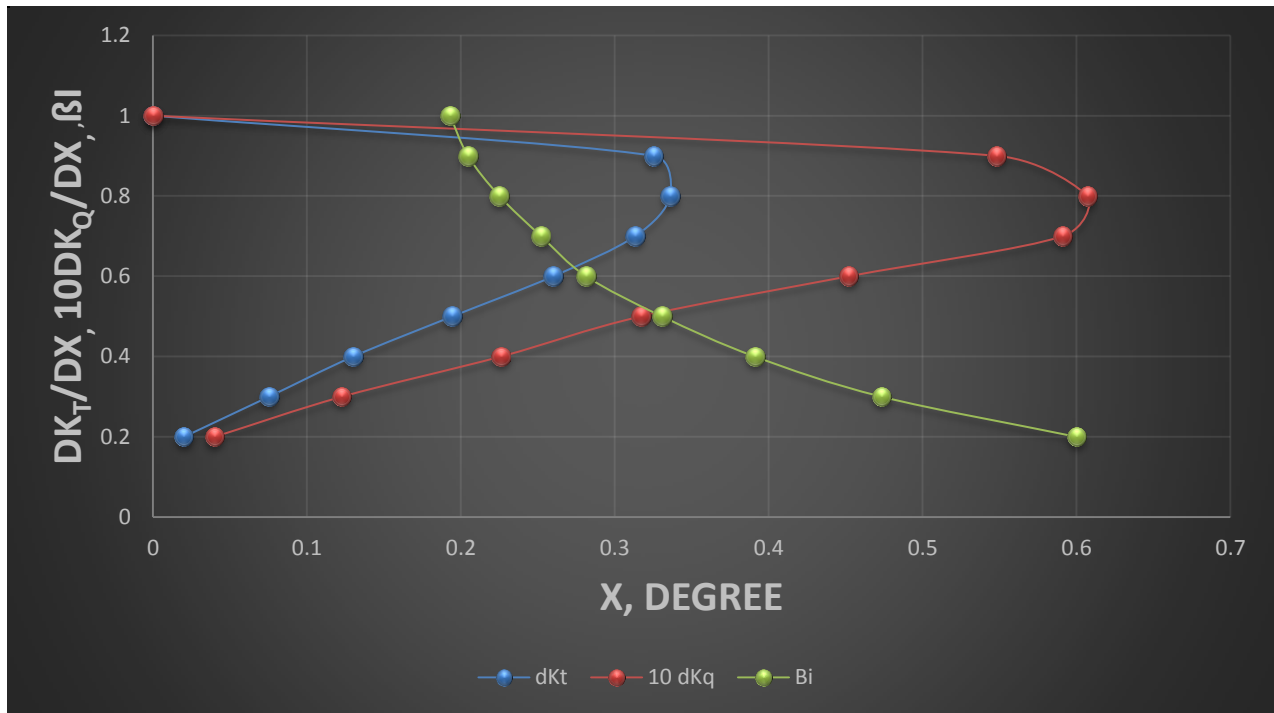


Figure 3.1: x vs dK_T/dx , dK_Q/dx , β_i

3.2.2 For Part load ($J=0.7$):

$x=.3$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A = 1\text{m/s}$

$J=V_A/nD$

So, $n = V_A/JD = \frac{1}{.7*0.3048} = 4.69$

$\tan\beta = V_A/(n\pi D x) = \frac{1}{4.69*3.1416*0.3048*.3} = .742$

$\beta=36.584 \text{ deg.}$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2) \cdot \frac{1-x}{x \sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .3458 - \frac{8 \cdot .3}{3 \cdot .3636 \cdot 57.2958} [\cos^{-1}\{e^{((-3/2) \cdot \frac{1-.3}{.3 \cdot \sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - 36.584) = 0$$

Solving for β_i ,

$$\beta_i = 47.2121 \text{ deg.}$$

Calculation for V_R :

$$V_R = V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)}$$

$$= 1 * \frac{\cos(47.2121 - 36.584)}{\sin(36.584)}$$

$$= 1.649 \text{ m/s}$$

Calculation for $\frac{dKT}{dx}$:

$$\frac{dKT}{dx} = z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos \beta_i (1 - \tan \beta_i * \tan \gamma)$$

$$\frac{dKT}{dx} = 3 * .3485 * \frac{.3636}{4 * 4.69^2 * .3048^2} * 1.649^2 * \cos(47.2121) (1 - \tan(47.2121) * \tan(.01314 / .2606))$$

$$= .0753$$

Calculation for $\frac{dKQ}{dx}$:

$$\frac{dKQ}{dx} = r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos \beta_i * (\tan \beta_i + \tan \gamma)$$

$$= (.3048 * .3/2) * 3 * .3485 * \frac{.3636}{4 * 4.69^2 * .3048^3} * 1.649^2 * \cos(47.2121) (\tan(47.2121) + \tan(.01314 / .2606))$$

$$= .014$$

$$x=.5$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A=1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n=V_A/JD=\frac{1}{.7*.3048}=4.69$$

$$\tan\beta=V_A/(\pi n D x)=\frac{1}{4.69*3.1416*.3048*.5}=.445$$

$$\beta=24 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2). \frac{1-x}{x \sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .2972 - \frac{8*.5}{3*.4392*57.2958} [\cos^{-1}\{e^{((-3/2). \frac{1-.5}{.5*\sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - 24) = 0$$

Solving for β_i ,

$$\beta_i=31.122 \text{ deg.}$$

Calculation for V_R :

$$\begin{aligned} V_R &= V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)} \\ &= 1 * \frac{\cos(31.122 - 24)}{\sin(24)} \\ &= 2.4396 \text{ m/s} \end{aligned}$$

Calculation for $\frac{dKT}{dx}$:

$$\begin{aligned} \frac{dKT}{dx} &= z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos \beta_i (1 - \tan \beta_i * \tan \gamma) \\ \frac{dKT}{dx} &= 3 * .2972 * \frac{.4392}{4*4.69^2*0.3048^2} * 2.4396^2 * \cos(31.122) (1 - \tan(31.122) * \tan(.01255/.2972)) \\ &= .3458 \end{aligned}$$

Calculation for $\frac{dKQ}{dx}$:

$$\begin{aligned} \frac{dKQ}{dx} &= r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos \beta_i * (\tan \beta_i + \tan \gamma) \\ &= (.3048 * .5/2) * 3 * .2972 * \frac{.4392}{4*4.69^2*0.3048^3} * 2.4396^2 * \cos(31.122) (\tan(31.122) + \tan(.01255/.2972)) \\ &= .0522 \end{aligned}$$

$$x=.7$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A=1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n=V_A/JD=\frac{1}{.7*.3048}=4.69$$

$$\tan\beta=V_A/(n\pi D x)=\frac{1}{4.69*3.1416*.3048*.7}=.318$$

$$\beta=17.646 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((z/2). \frac{1-x}{x \sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .2421 - \frac{8*.7}{3*.4622*57.2958} [\cos^{-1}\{e^{((-3/2). \frac{1-.7}{.7 \sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - 17.646) = 0$$

$$\text{Solving for } \beta_i, \quad \beta_i=23.248 \text{ deg.}$$

Calculation for V_R :

$$V_R = V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)}$$

$$= 1 * \frac{\cos(23.248 - 17.646)}{\sin(17.646)}$$

$$= 3.315 \text{ m/s}$$

Calculation for $\frac{dKT}{dx}$:

$$\frac{dKT}{dx} = z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos \beta_i (1 - \tan \beta_i * \tan \gamma)$$

$$\frac{dKT}{dx} = 3 * .2421 * \frac{.4622}{4*4.69^2*.3048^2} * 3.315^2 * \cos(23.248) (1 - \tan(23.248) * \tan(.01106/.2421))$$

$$=.4145$$

Calculation for $\frac{dKQ}{dx}$:

$$\frac{dKQ}{dx} = r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos \beta_i * (\tan \beta_i + \tan \gamma)$$

$$= (.3048*7/2) * 3 * .2421 * \frac{.4622}{4*4.69^2*.3048^3} * 3.315^2 * \cos(23.248) (\tan(23.248) + \tan(.01106/.2421))$$

$$=.0624$$

$$x=.9$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A=1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n=V_A/JD=\frac{1}{.7*.3048}=4.69$$

$$\tan\beta=V_A/(n\pi D x)=\frac{1}{4.69*3.1416*.3048*.9}=.2474$$

$$\beta=13.897 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((z/2). \frac{1-x}{x \sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .1798 - \frac{8*.9}{3*.3613*57.2958} [\cos^{-1}\{e^{((-3/2). \frac{1-.9}{.9*\sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - 13.897) = 0$$

Solving for β_i , $\beta_i=19.014 \text{ deg.}$

Calculation for V_R :

$$V_R = V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)}$$

$$= 1 * \frac{\cos(19.014 - 13.897)}{\sin(13.897)}$$

$$= 4.147 \text{ m/s}$$

Calculation for $\frac{dKT}{dx}$:

$$\frac{dKT}{dx} = z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos \beta_i (1 - \tan \beta_i * \tan \gamma)$$

$$\frac{dKT}{dx} = 3 * .1792 * \frac{.3613}{4*4.69^2*.3048^2} * 4.147^2 * \cos(19.014) (1 - \tan(19.014) * \tan(.00978/.1798))$$

$$=.3875$$

Calculation for $\frac{dKQ}{dx}$:

$$\frac{dKQ}{dx} = r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos \beta_i * (\tan \beta_i + \tan \gamma)$$

$$= (.3048 * .9/2) * 3 * .1792 * \frac{.3613}{4*4.69^2*.3048^3} * 4.147^2 * \cos(19.014) (\tan(19.014) + \tan(.00978/.1798))$$

$$=.0603$$

3.2.2.1 Table for K_T , K_Q , π for $J=0.7$

X	dK _T /dx	SM	Product	dK _Q /dx	SM	Product
0.2	0.03265	1	0.03265	0.00669	1	0.00669
0.3	0.085	4	0.34	0.014	4	0.056
0.4	0.21733	2	0.43466	0.03414	2	0.06828
0.5	0.345	4	1.38	0.0522	4	0.2088
0.6	0.39759	2	0.79518	0.05859	2	0.11718
0.7	0.4145	4	1.658	0.0624	4	0.2496
0.8	0.4077	2	0.8154	0.063107	2	0.126214
0.9	0.3875	4	1.55	0.0603	4	0.2412
1	0	1	0	0	1	0
			7.00589			1.073964

$$K_T = 0.233529667$$

$$K_Q = 0.0357988$$

$$\eta = 0.726759549$$

3.2.2.2 Graph for x Vs dK_T/dx , dK_Q/dx , β_i :

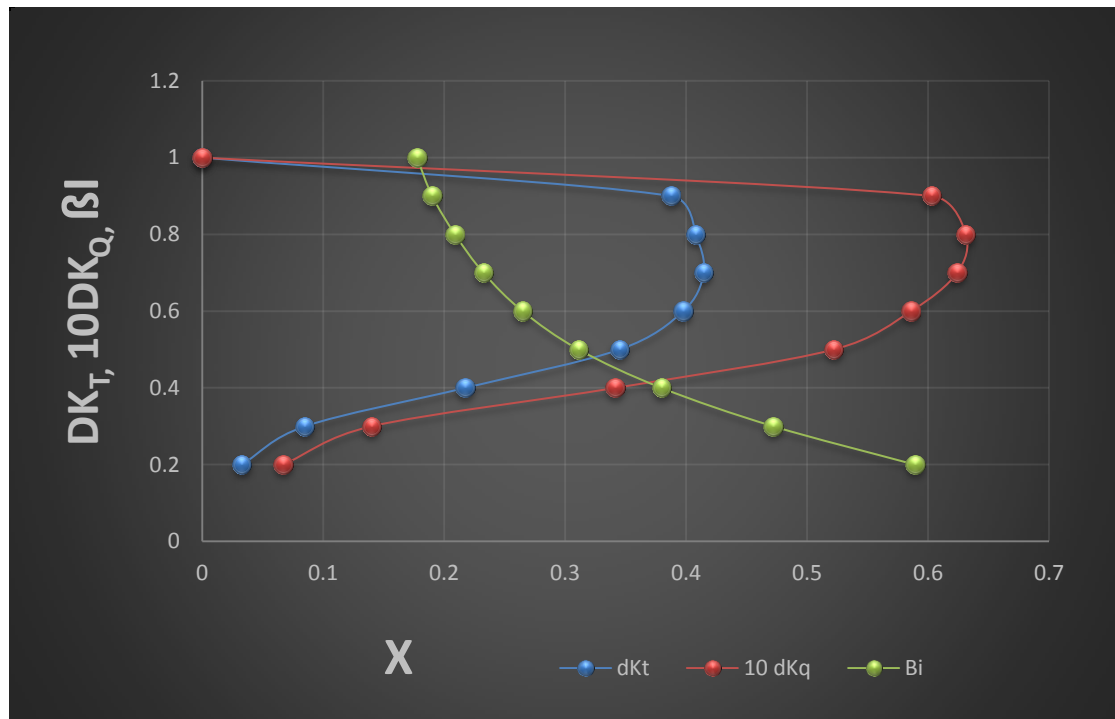


Figure 3.2: x Vs dK_T/dx , dK_Q/dx , β_i

3.2.3 : For Over load (J=1):

$x=.3$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A=1\text{m/s}$

$J=V_A/nD$

So, $n = V_A/JD = \frac{1}{.1 \cdot .3048} = 3.28$

$\tan\beta = V_A/(n\pi D x) = \frac{1}{3.28 \cdot 3.1416 \cdot .3048 \cdot .3} = 1.061$

$\beta=46.703 \text{ deg.}$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2) \cdot \frac{1-x}{x \sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .1406 - \frac{8 \cdot .3}{3 \cdot .3636 \cdot 57.2958} [\cos^{-1}\{e^{((-3/2) \cdot \frac{1-.3}{.3 \cdot \sin \beta_i})}\}] * \sin \beta_i \cdot \tan(\beta_i - 46.703) = 0$$

Solving for β_i ,

$$\beta_i = 46.759 \text{ deg.}$$

Calculation for V_R :

$$\begin{aligned} V_R &= V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)} \\ &= 1 * \frac{\cos(46.759 - 46.703)}{\sin(46.703)} \\ &= 1.458 \text{ m/s} \end{aligned}$$

Calculation for $\frac{dKT}{dx}$:

$$\begin{aligned} \frac{dKT}{dx} &= z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos \beta_i (1 - \tan \beta_i * \tan \gamma) \\ \frac{dKT}{dx} &= 3 * .1406 * \frac{.3636}{4 * 3.28^2 * .3048^2} * 1.458^2 * \cos(46.759) (1 - \tan(46.759) * \tan(.01144 / .1406)) \\ &= .0503 \end{aligned}$$

Calculation for $\frac{dKQ}{dx}$

$$\begin{aligned} \frac{dKQ}{dx} &= r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos \beta_i * (\tan \beta_i + \tan \gamma) \\ &= (.3048 * .3/2) * 3 * .1406 * \frac{.3636}{4 * 3.28^2 * .3048^3} * 1.458^2 * \cos(46.759) (\tan(46.759) + \tan(.01144 / .1406)) \\ &= .0089 \end{aligned}$$

$$x=.5$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A=1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n=V_A/JD=\frac{1}{.1*3048}=3.28$$

$$\tan\beta=V_A/(n\pi D x)=\frac{1}{3.28*3.1416*3048*.5}=.6368$$

$$\beta=32.488 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2). \frac{1-x}{x \sin \beta_i})}\}] * \sin \beta_i \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .1311 - \frac{8*.5}{3*4392*57.2958} [\cos^{-1}\{e^{((-3/2). \frac{1-.5}{.5*\sin \beta_i})}\}] * \sin \beta_i \tan(\beta_i - 32.488) = 0$$

Solving for β_i ,

$$\beta_i=32.548 \text{ deg.}$$

Calculation for V_R :

$$\begin{aligned} V_R &= V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)} \\ &= 1 * \frac{\cos(32.548 - 32.488)}{\sin(24)} \\ &= 1.862 \text{ m/s} \end{aligned}$$

Calculation for $\frac{dKT}{dx}$:

$$\begin{aligned} \frac{dKT}{dx} &= z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos \beta_i (1 - \tan \beta_i * \tan \gamma) \\ \frac{dKT}{dx} &= 3 * .1311 * \frac{.4392}{4*3.28^2*0.3048^2} * 1.862^2 * \cos(32.548) (1 - \tan(32.548) * \tan(.01085/.1311)) \\ &= .1033 \end{aligned}$$

Calculation for $\frac{dKQ}{dx}$:

$$\begin{aligned} \frac{dKQ}{dx} &= r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos \beta_i * (\tan \beta_i + \tan \gamma) \\ &= (.3048 * .5/2) * 3 * .1311 * \frac{.4392}{4*3.28^2*0.3048^3} * 1.862^2 * \cos(32.548) (\tan(32.548) + \tan(.01085/.1311)) \\ &= .0202 \end{aligned}$$

$$x=.7$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A = 1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n = V_A/JD = \frac{1}{.1*3048} = 3.28$$

$$\tan\beta = V_A/(n\pi D x) = \frac{1}{3.28*3.1416*3048*.7} = .4575$$

$$\beta = 24.45 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2)(\frac{1-x}{x\sin\beta_i})}\}]] * \sin\beta_i \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .1204 - \frac{8*.7}{3*.4622*57.2958} [\cos^{-1}\{e^{((-3/2)(\frac{1-.7}{.7*\sin\beta_i})}\}]] * \sin\beta_i \tan(\beta_i - 24.45) = 0$$

Solving for β_i , $\beta_i = 24.503 \text{ deg.}$

Calculation for V_R :

$$V_R = V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)}$$

$$= 1 * \frac{\cos(24.503 - 24.45)}{\sin(24.45)}$$

$$= 2.42 \text{ m/s}$$

Calculation for $\frac{dKT}{dx}$:

$$\frac{dKT}{dx} = z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos\beta_i (1 - \tan\beta_i * \tan\gamma)$$

$$\frac{dKT}{dx} = 3 * .1204 * \frac{.4622}{4 * 3.28^2 * .3048^2} * 2.42^2 * \cos(24.503) (1 - \tan(24.503) * \tan(.01006/.1204))$$

$$= .1483$$

Calculation for $\frac{dKQ}{dx}$:

$$\frac{dKQ}{dx} = r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos\beta_i * (\tan\beta_i + \tan\gamma)$$

$$= (.3048 * 7/2) * 3 * .1204 * \frac{.4622}{4 * 3.28^2 * .3048^3} * 2.42^2 * \cos(24.503) (\tan(24.503) + \tan(.01006/.1204))$$

$$= .0624$$

$$x=.9$$

No of blades, $z=3$

Propeller Diameter, $D=0.3048\text{m}$

Speed of advance, $V_A = 1\text{m/s}$

$$J=V_A/nD$$

$$\text{So, } n = V_A/JD = \frac{1}{.1*.3048} = 3.28$$

$$\tan\beta = V_A/(n\pi D x) = \frac{1}{3.28*3.1416*.3048*.9} = .3537$$

$$\beta = 19.4775 \text{ deg.}$$

Calculation for β_i :

$$C_L - \frac{8x}{z(\frac{C}{D})} [\cos^{-1}\{e^{((-z/2). \frac{1-x}{x \sin \beta_i})}\}] * \sin \beta_i \tan(\beta_i - \beta) = 0$$

$$\text{Or, } .1061 - \frac{8*.9}{3*.3613*57.2958} [\cos^{-1}\{e^{((-3/2). \frac{1-.9}{.9*\sin \beta_i})}\}] * \sin \beta_i \tan(\beta_i - 19.4775) = 0$$

Solving for β_i ,

$$\beta_i = 21.927 \text{ deg.}$$

Calculation for V_R :

$$V_R = V_A * \frac{\cos(\beta_i - \beta)}{\sin(\beta)}$$

$$= 1 * \frac{\cos(21.927 - 19.4775)}{\sin(19.4775)}$$

$$= 2.675 \text{ m/s}$$

Calculation for $\frac{dKT}{dx}$:

$$\frac{dKT}{dx} = z * C_L * \frac{C/D}{4n^2 D^2} * V_R^2 * \cos \beta_i (1 - \tan \beta_i * \tan \gamma)$$

$$\frac{dKT}{dx} = 3 * .1061 * \frac{.3613}{4*3.28^2*.3048^2} * 2.675^2 * \cos(21.927) (1 - \tan(21.927) * \tan(.00986/.1061))$$

$$= .1458$$

Calculation for $\frac{dKQ}{dx}$:

$$\frac{dKQ}{dx} = r * z * C_L * \frac{C/D}{4n^2 D^3} * V_R^2 * \cos \beta_i * (\tan \beta_i + \tan \gamma)$$

$$= (.3048 * .9/2) * 3 * .1061 * \frac{.3613}{4*3.28^2*.3048^3} * 2.675^2 * \cos(21.927) (\tan(21.927) + \tan(.00986/.1061))$$

$$= .0603$$

3.2.3.1 Table for K_T , K_Q , η for $J=1$

x	dK_T/dx	SM	Product	dK_Q/dx	SM	Product
0.2	0.029254	1	0.029254	0.00478	1	0.00478
0.3	0.0503	4	0.2012	0.0089	4	0.0356
0.4	0.0723	2	0.1446	0.0144	2	0.0288
0.5	0.1033	4	0.4132	0.0202	4	0.0808
0.6	0.128	2	0.256	0.02478	2	0.04956
0.7	0.1483	4	0.5932	0.029	4	0.116
0.8	0.1583	2	0.3166	0.03331	2	0.06662
0.9	0.1458	4	0.5832	0.0351	4	0.1404
1	0	1	0	0	1	0
			2.537254			0.52256

$$K_T = 0.084575$$

$$K_Q = 0.017419$$

$$\eta = 0.772764$$

3.2.3.2 Graph for x Vs dK_T/dx , dK_Q/dx , β_i

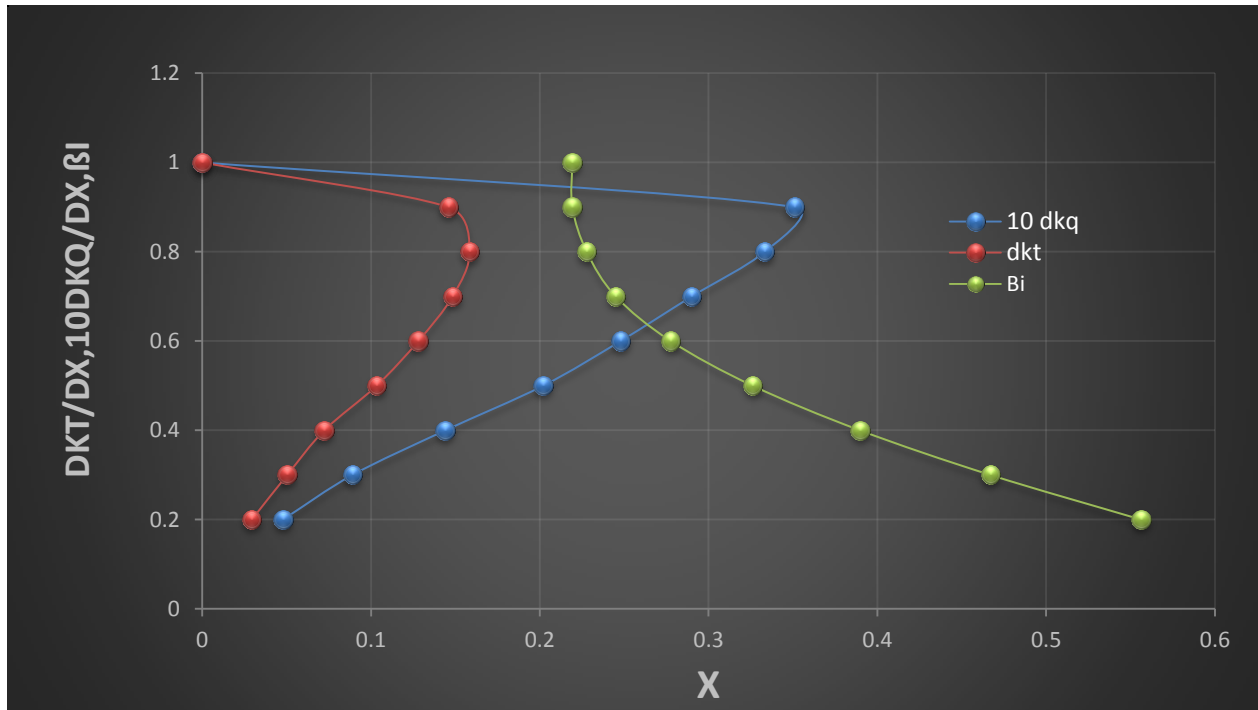


Figure 3.3: x Vs dK_T/dx , dK_Q/dx , β_i

3.2.4 Combined dK_T/dx for $J=1,.7,.833$

	J=.1	J=.7	J=.833
x	dK_T/dx	dK_T/dx	dK_T/dx
0.2	0.029254	0.03265	0.02
0.3	0.0503	0.085	0.0753
0.4	0.0723	0.21733	0.13
0.5	0.1033	0.345	0.1944
0.6	0.128	0.39759	0.26
0.7	0.1483	0.4145	0.3132
0.8	0.1583	0.4077	0.336
0.9	0.1458	0.3875	0.3252
1	0	0	0

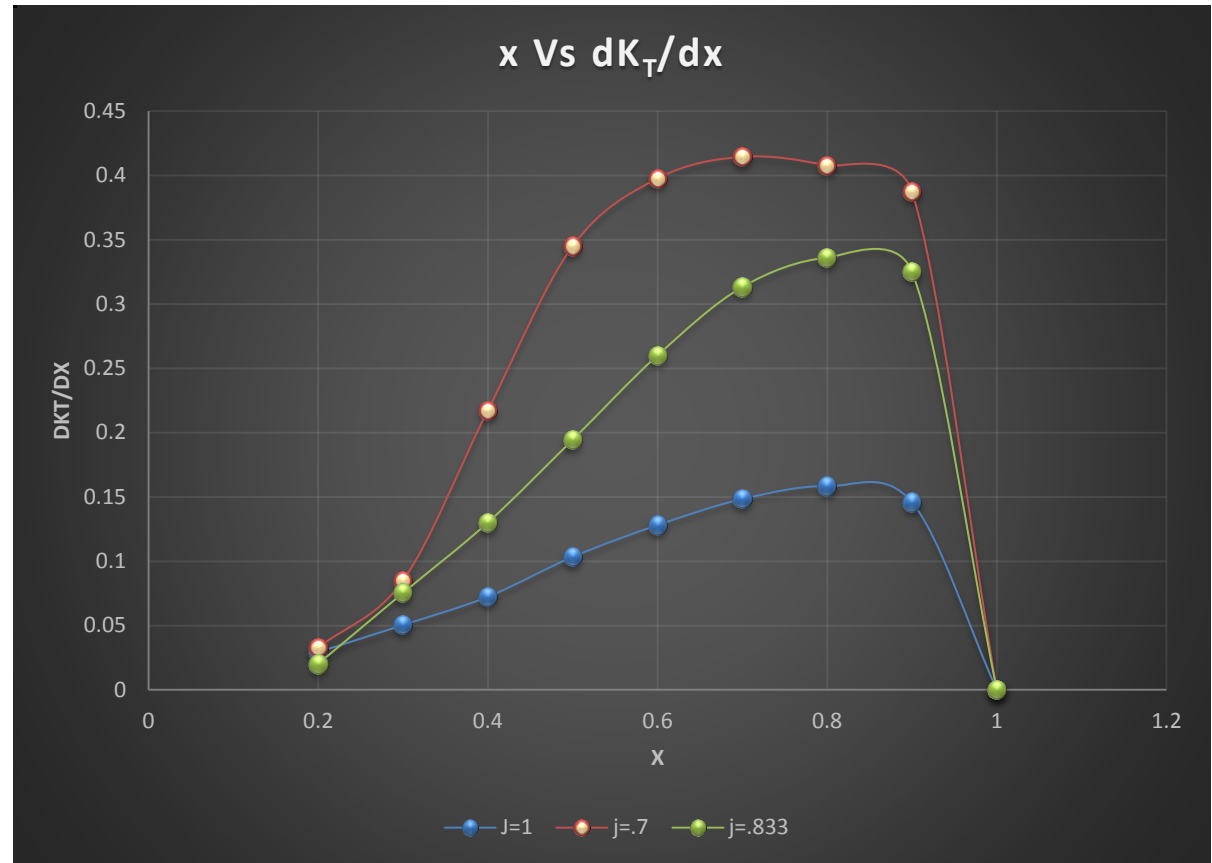


Figure 3.4: x Vs dK_T/dx

3.2.5 Combined dK_Q/dx for $J=1, .7, .833$

	J=1	J=.7	J=.833
x	dK_Q/dx	dK_Q/dx	dK_Q/dx
0.2	0.00478	0.00669	0.004
0.3	0.0089	0.014	0.01227
0.4	0.0144	0.03414	0.0226
0.5	0.0202	0.0522	0.0317
0.6	0.02478	0.05859	0.0452
0.7	0.029	0.0624	0.0591
0.8	0.03331	0.063107	0.0607
0.9	0.0351	0.0603	0.0548
1	0	0	0

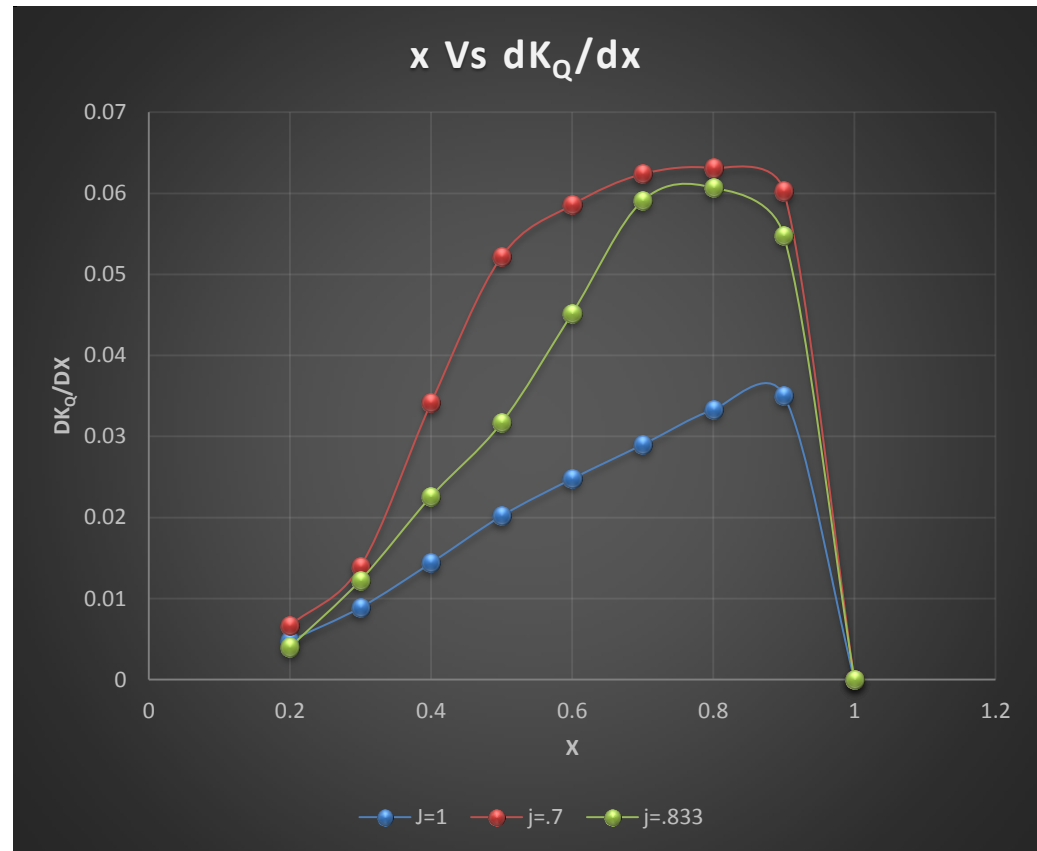


Figure 3.4: x Vs dK_Q/dx

3.2.6 Comparison Between Calculated and Burrill Analysis

J=.833	Present calculation			From Burrill analysis	
x	β_i	dK_T/dx	dK_Q/dx	dK_T/dx	dK_Q/dx
0.3	47.34	0.0753	0.01227	0.1154	0.0173
0.5	33.07	0.1944	0.0317	0.2124	0.0339
0.7	25.208	0.3132	0.0591	0.314	0.0524
0.9	20.46	0.3252	0.0548	0.3093	0.0543

5

J=1	Present calculation			From Burrill analysis	
x	β_i	dK_T/dx	dK_Q/dx	dK_T/dx	dK_Q/dx
0.3	46.759	0.0503	0.0089	0.0506	0.0104
0.5	32.548	0.1033	0.0202	0.1013	0.0215
0.7	24.503	0.1483	0.0624	0.1589	0.0359
0.9	21.927	0.1458	0.0603	0.1659	0.0408

J=.7	Present calculation			From Burrill analysis	
x	β_i	dK_T/dx	dK_Q/dx	dK_T/dx	dK_Q/dx
0.3	47.2121	0.085	0.014	0.1684	0.0213
0.5	31.122	0.3458	0.0522	0.3158	0.0434
0.7	23.248	0.4145	0.0624	0.4492	0.0678
0.9	19.014	0.3875	0.0603	0.4071	0.0678

Open water Characteristics						
J	K_T	Calculated K_T	K_Q	Calculated K_Q	η	Calculated η
0.7	0.222	0.233529667	0.0386	0.0357988	0.6415	0.726759549
0.833	0.1537	0.1701	0.028727	0.0297	0.6458	0.758
1	0.0629	0.084575133	0.0144	0.017418667	0.6931	0.77276403

3.2.7 Graph of Advance co-efficient Vs Efficiency

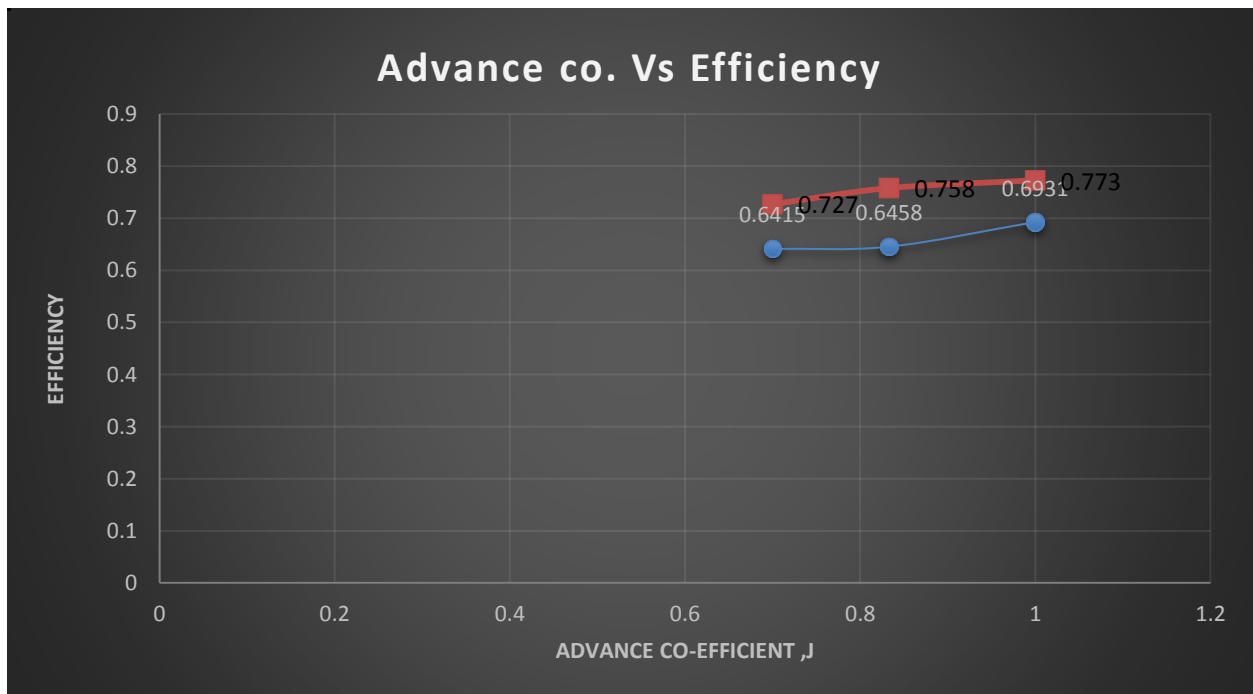


Figure 3.5: Advance co-efficient Vs Efficiency

CHAPTER 4

PRESSURE CALCULATION

4.1 Pressure Distribution Calculation

Now, using C_L value we obtained through iteration is used to calculate pressure distribution around a blade section.

A hydrofoil Section comprises NACA a=0.8 mean line and NACA 66 (modified) thickness form.

For camber 6.79% the relevant velocity data is provided below:

	v/V (for NACA 66(mod) thickness form)	$\Delta v_a/V$ (for incidence corresponding to $CL=1.0$ & NACA 66(mod) thickness	$\Delta v/V$ (for NACA a=.8 meanline & corresponding to $Cl_i=1$)
0	0	2.987	0.278
10	1.093	0.478	0.278
20	1.111	0.323	0.278
30	1.121	0.249	0.278
40	1.128	0.198	0.278
50	1.127	0.161	0.278
60	1.099	0.127	0.278
70	1.064	0.097	0.278
80	1.023	0.071	0.278
90	0.974	0.045	0.139
100	0	0	0

Velocity at any point is given by:

$$v/V(\text{total}) = v/V (+\text{or-}) \Delta v_a/V (+\text{or-}) \Delta v/V$$

v/V = velocity due to thickness form set at 0 incidence and positive for back and face

$\Delta v_a/V$ = the increment of velocity due to angle of incidence positive for back and negative for face

$\Delta v/V$ = the increment of velocity due to the meanline and positive for back and negative for face

Now the expression for pressure co-efficient:

$$\Delta p/q = 1 - [v/V(\text{total})]^2$$

$$= 1 - [v/V (+\text{or-}) \Delta v_a/V (+\text{or-}) \Delta v/V]^2$$

$$v/V(\text{Total})\mathbf{Back} = [v/V + \Delta v_a/V + \Delta v/V]$$

$$v/V(\text{Total})\mathbf{Face} = [v/V - \Delta v_a/V - \Delta v/V]$$

$$\Delta p/q = 1 - [v/V(\text{total})\mathbf{Back}]^2$$

$$\Delta p/q = 1 - [v/V(\text{total})\mathbf{Face}]^2$$

4.2 Calculation table for $C_L(J = .833)$

For $x=.5$, $C_L=.2606$, $\Delta v/V$ is multiplied by $2.18/6.79=0.321$. Lift developed by camber= $1*0.321=.321$

Lift to be developed= $0.2606-0.321=-0.0604$. $\Delta v_a/V$ multiplied by -0.0604

for $x=.5$ $C_L = .2606$											
	C_x (From L.E)	v/V	$\Delta v_a/V$	$\Delta v/V$	v/V (Total) back	$(\Delta p/q)$ back	v/V (Total) face	$(\Delta p/q)$ face	$(\Delta p/q)$ face - $(\Delta p/q)$ back	S.M	Product
1	0	0	-0.18041	0.089238	-0.0911768	0.99168679	0.091177	0.99168679	0	1	0
2	0.1	1.093	-0.02887	0.089238	1.1533668	-0.330255	1.032633	-0.0663313	0.263924	4	1.055695
3	0.2	1.111	-0.01951	0.089238	1.1807288	-0.3941205	1.041271	-0.0842457	0.309875	2	0.61975
4	0.3	1.121	-0.01504	0.089238	1.1951984	-0.4284992	1.046802	-0.0957936	0.332706	4	1.330823
5	0.4	1.128	-0.01196	0.089238	1.2052788	-0.452697	1.050721	-0.104015	0.348682	2	0.697364
6	0.5	1.127	-0.00972	0.089238	1.2065136	-0.4556751	1.047486	-0.0972278	0.358447	4	1.433789
7	0.6	1.099	-0.00767	0.089238	1.1805672	-0.3937389	1.017433	-0.0351695	0.358569	2	0.717139
8	0.7	1.064	-0.00586	0.089238	1.1473792	-0.316479	0.980621	0.03838285	0.354862	4	1.419448
9	0.8	1.023	-0.00429	0.089238	1.1079496	-0.2275523	0.93805	0.12006145	0.347614	2	0.695228
10	0.9	0.974	-0.00272	0.044619	1.015901	-0.0320548	0.932099	0.13119145	0.163246	4	0.652985
11	1	0	0	0	0	1	0	1	0	1	0
Lift co-efficient, $C_L = 0.287407$											8.622219

For $x=.7$, $C_L=.1865$, $\Delta v/V$ is multiplied by $2/6.79=0.295$. Lift developed by camber= $1*0.295=.295$

Lift to be developed= $0.1865-0.295=-0.1085$. $\Delta v_a/V$ multiplied by $-.1085$

for $x=.7$ $C_L=.1865$											
	C_x (From L.E)	v/V	$\Delta v_a/V$	$\Delta v/V$	v/V (Total) back	$(\Delta p/q)$ back	v/V (Total) face	$(\Delta p/q)$ face	$(\Delta p/q)$ face - $(\Delta p/q)$ back	S.M	Product
1	0	0	-0.32409	0.08201	-0.2420795	0.94139752	0.24208	0.94139752	0	1	0
2	0.1	1.093	-0.05186	0.08201	1.123147	-0.2614592	1.062853	-0.1296565	0.131803	4	0.527211
3	0.2	1.111	-0.03505	0.08201	1.1579645	-0.3408818	1.064036	-0.1321715	0.20871	2	0.41742
4	0.3	1.121	-0.02702	0.08201	1.1759935	-0.3829607	1.066007	-0.1363699	0.246591	4	0.986363
5	0.4	1.128	-0.02148	0.08201	1.188527	-0.4125964	1.067473	-0.1394986	0.273098	2	0.546196
6	0.5	1.127	-0.01747	0.08201	1.1915415	-0.4197711	1.062459	-0.1288181	0.290953	4	1.163812
7	0.6	1.099	-0.01378	0.08201	1.1672305	-0.362427	1.03077	-0.0624858	0.299941	2	0.599883
8	0.7	1.064	-0.01052	0.08201	1.1354855	-0.2893273	0.992515	0.01491497	0.304242	4	1.216969
9	0.8	1.023	-0.0077	0.08201	1.0973065	-0.2040816	0.948694	0.09998064	0.304062	2	0.608124
10	0.9	0.974	-0.00488	0.041005	1.0101225	-0.0203475	0.937878	0.12038579	0.140733	4	0.562933
11	1	0	0	0	0	1	0	1	0	1	0
											6.628912

Lift co-efficient, $C_L = 0.220964$

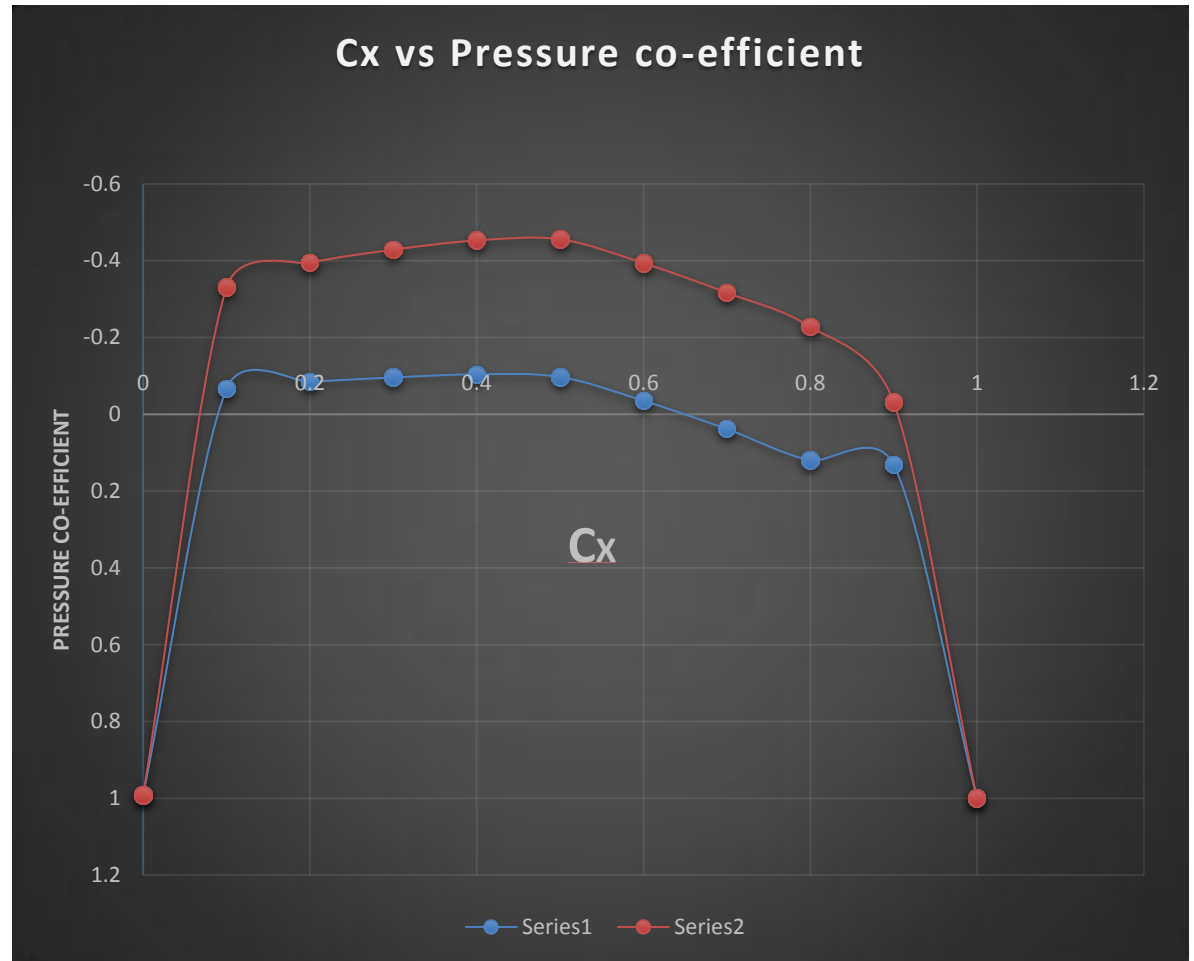
For $x=.9$, $CL=.1482$, $\Delta v/V$ is multiplied by $1.82/6.79=0.268$. Lift developed by camber= $1*0.268=.268$

Lift to be developed= $0.1482-0.268=-0.1198$. $\Delta va/V$ multiplied by -0.1198

for $x=.9$ $CL = .1482$											
	Cx (From L.E)	v/V	$\Delta va/V$	$\Delta v/V$	v/V (Total) back	($\Delta p/q$) back	v/V (Total) face	($\Delta p/q$) face	($\Delta p/q$) face - ($\Delta p/q$) back	S.M	Product
1	0	0	-0.35784	0.074504	-0.2833386	0.91971924	0.283339	0.91971924	0	1	0
2	0.1	1.093	-0.05726	0.074504	1.1102396	-0.232632	1.07576	-0.1572604	0.075372	4	0.301486
3	0.2	1.111	-0.0387	0.074504	1.1468086	-0.31517	1.075191	-0.1560365	0.159133	2	0.318267
4	0.3	1.121	-0.02983	0.074504	1.1656738	-0.3587954	1.076326	-0.1584781	0.200317	4	0.801269
5	0.4	1.128	-0.02372	0.074504	1.1787836	-0.3895308	1.077216	-0.1603952	0.229136	2	0.458271
6	0.5	1.127	-0.01929	0.074504	1.1822162	-0.3976351	1.071784	-0.1487205	0.248915	4	0.995659
7	0.6	1.099	-0.01521	0.074504	1.1582894	-0.3416343	1.039711	-0.0809981	0.260636	2	0.521272
8	0.7	1.064	-0.01162	0.074504	1.1268834	-0.2698662	1.001117	-0.0022344	0.267632	4	1.070527
9	0.8	1.023	-0.00851	0.074504	1.0889982	-0.1859171	0.957002	0.08414755	0.270065	2	0.540129
10	0.9	0.974	-0.00539	0.037252	1.005861	-0.0117564	0.942139	0.1123741	0.12413	4	0.496522
11	1	0	0	0	0	1	0	1	0	1	0
Lift co-efficient, $C_L = 0.183447$											5.503402

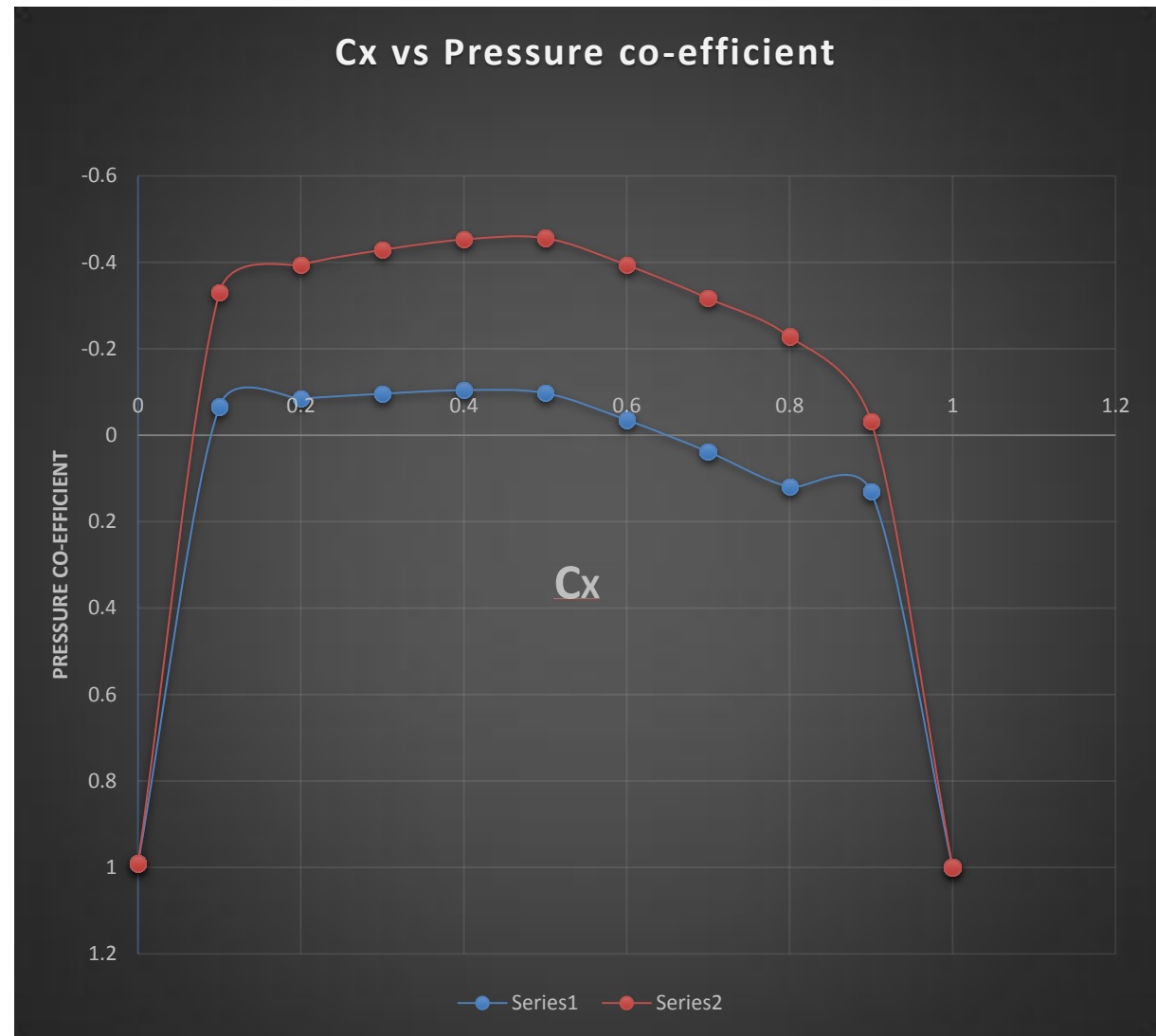
4.2.1 Graph of Cx vs Pressure co-efficient for 2.18% camber:

Graph for 2.18% camber		
	face	back
0	0.99168679	0.99168679
0.1	-0.0663313	-0.330255
0.2	-0.0842457	-0.3941205
0.3	-0.0957936	-0.4284992
0.4	-0.104015	-0.452697
0.5	-0.0972278	-0.4556751
0.6	-0.0351695	-0.3937389
0.7	0.03838285	-0.316479
0.8	0.12006145	-0.2275523
0.9	0.13119145	-0.0320548
1	1	1



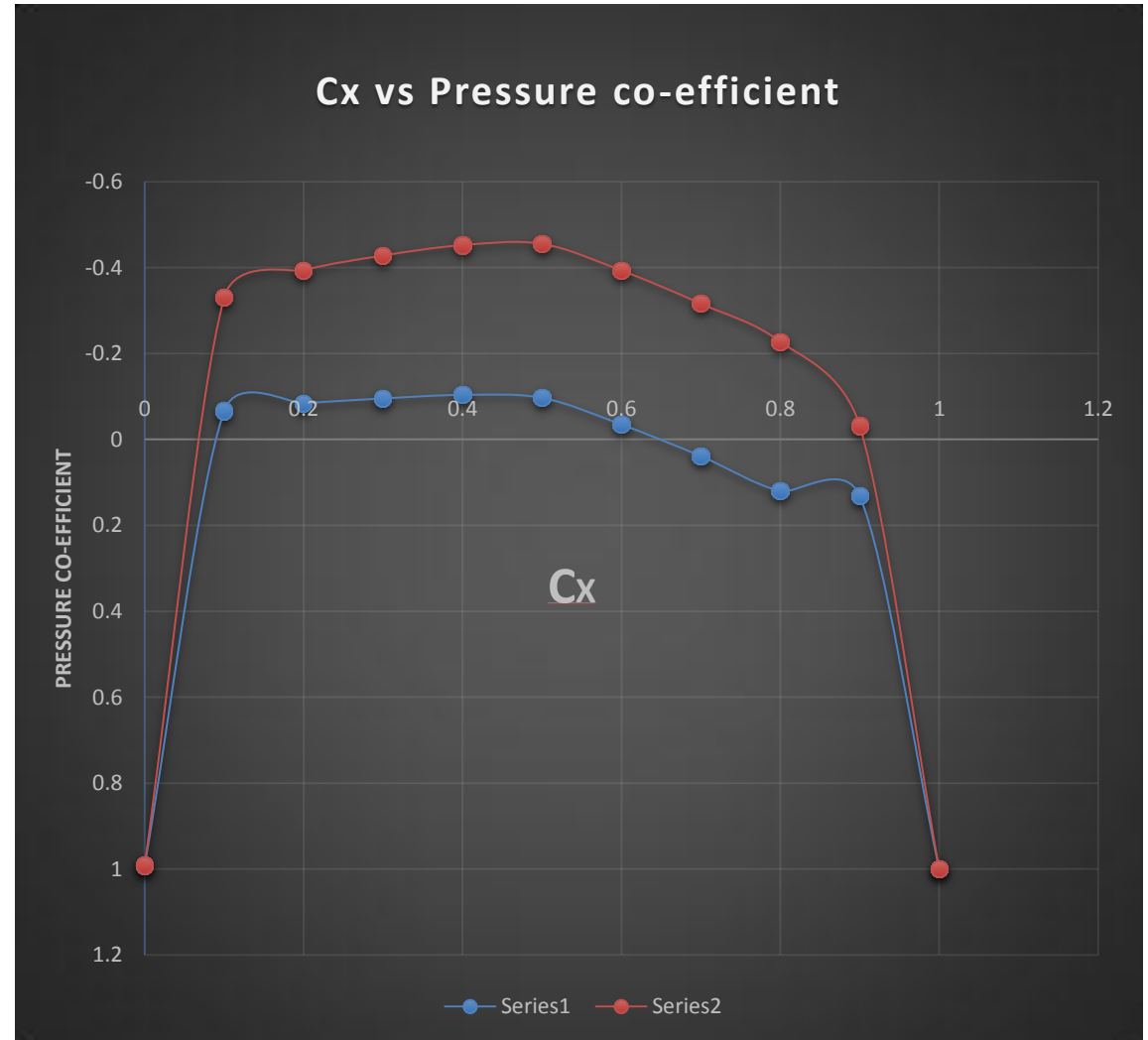
4.2.2 Graph of Cx vs Pressure co-efficient For 2% camber :

Graph for 2% camber		
	face	back
0	0.99422365	0.99422365
0.1	-0.0738684	-0.3218648
0.2	-0.0942765	-0.3827956
0.3	-0.1070451	-0.4157142
0.4	-0.1161166	-0.4388861
0.5	-0.1098768	-0.4411837
0.6	-0.0479787	-0.378961
0.7	0.02559154	-0.3016042
0.8	0.10745518	-0.2127606
0.9	0.12507267	-0.0254084
1	1	1



4.2.3 Graph of Cx vs Pressure co-efficient for 1.82% camber

	face	back
0	0.96429812	0.96429812
0.1	-0.1249903	-0.2664
0.2	-0.1341922	-0.3386846
0.3	-0.1416021	-0.3772013
0.4	-0.1469545	-0.4043207
0.5	-0.1378408	-0.4096903
0.6	-0.0726691	-0.3509473
0.7	0.0038934	-0.2767838
0.8	0.08843678	-0.1908087
0.9	0.11505153	-0.014619
1	1	1



4.3 Calculation table for C_L (J = .7)

For $x=.5$, $CL=.1311$, $\Delta v/V$ is multiplied by $2.18/6.79=0.321$. Lift developed by camber= $1*0.321=.321$

Lift to be developed= $0.1311-0.321=-0.1899$. $\Delta va/V$ multiplied by -0.1899

for $x=.5$ $CL=.1311$											
	Cx (From L.E)	v/V	$\Delta va/V$	$\Delta v/V$	v/V (Total) back	($\Delta p/q$) back	v/V (Total) face	($\Delta p/q$) face	($\Delta p/q$) face - ($\Delta p/q$) back	S.M	Product
1	0	0	-0.56723	0.089238	-0.4779933	0.77152241	0.477993	0.77152241	0	1	0
2	0.1	1.093	-0.09077	0.089238	1.0914658	-0.1912976	1.094534	-0.1980051	-0.00671	4	-0.02683
3	0.2	1.111	-0.06134	0.089238	1.1389003	-0.2970939	1.0831	-0.173105	0.123989	2	0.247978
4	0.3	1.121	-0.04729	0.089238	1.1629529	-0.3524594	1.079047	-0.1643426	0.188117	4	0.752467
5	0.4	1.128	-0.0376	0.089238	1.1796378	-0.3915453	1.076362	-0.1585556	0.23299	2	0.46598
6	0.5	1.127	-0.03057	0.089238	1.1856641	-0.4057994	1.068336	-0.1413416	0.264458	4	1.057831
7	0.6	1.099	-0.02412	0.089238	1.1641207	-0.355177	1.033879	-0.0689064	0.286271	2	0.572541
8	0.7	1.064	-0.01842	0.089238	1.1348177	-0.2878112	0.993182	0.01358892	0.3014	4	1.205601
9	0.8	1.023	-0.01348	0.089238	1.0987551	-0.2072628	0.947245	0.1027271	0.30999	2	0.61998
10	0.9	0.974	-0.00855	0.044619	1.0100735	-0.0202485	0.937927	0.12029388	0.140542	4	0.562169
11	1	0	0	0	0	1	0	1	0	1	0

Lift co-efficient, C_L =0.181924

5.457716

For $x=.7$, $CL=.1204$, $\Delta v/V$ is multiplied by $2/6.79=0.295$. Lift developed by camber= $1*0.295=.295$

Lift to be developed= $0.1204-0.295=-0.1746$. $\Delta v_a/V$ multiplied by $-.1746$

for $x=.7$ $CL=.1204$											
	Cx (From L.E)	v/V	$\Delta v_a/V$	$\Delta v/V$	v/V (Total) back	($\Delta p/q$) back	v/V (Total) face	($\Delta p/q$) face	($\Delta p/q$) face - ($\Delta p/q$) back	S.M	Product
1	0	0	-0.52153	0.08201	-0.4395202	0.80682199	0.43952	0.80682199	0	1	0
2	0.1	1.093	-0.08346	0.08201	1.0915512	-0.191484	1.094449	-0.1978182	-0.00633	4	-0.02534
3	0.2	1.111	-0.0564	0.08201	1.1366142	-0.2918918	1.085386	-0.1780623	0.11383	2	0.227659
4	0.3	1.121	-0.04348	0.08201	1.1595346	-0.3445205	1.082465	-0.1717313	0.172789	4	0.691157
5	0.4	1.128	-0.03457	0.08201	1.1754392	-0.3816573	1.080561	-0.1676116	0.214046	2	0.428091
6	0.5	1.127	-0.02811	0.08201	1.1808994	-0.3945234	1.073101	-0.1515449	0.242978	4	0.971914
7	0.6	1.099	-0.02217	0.08201	1.1588358	-0.3429004	1.039164	-0.0798622	0.263038	2	0.526076
8	0.7	1.064	-0.01694	0.08201	1.1290738	-0.2748076	0.998926	0.00214645	0.276954	4	1.107816
9	0.8	1.023	-0.0124	0.08201	1.0926134	-0.193804	0.953387	0.09105399	0.284858	2	0.569716
10	0.9	0.974	-0.00786	0.041005	1.007148	-0.0143471	0.940852	0.11479751	0.129145	4	0.516578
11	1	0	0	0	0	1	0	1	0	1	0
Lift co-efficient, C_L =0.167122											5.013672

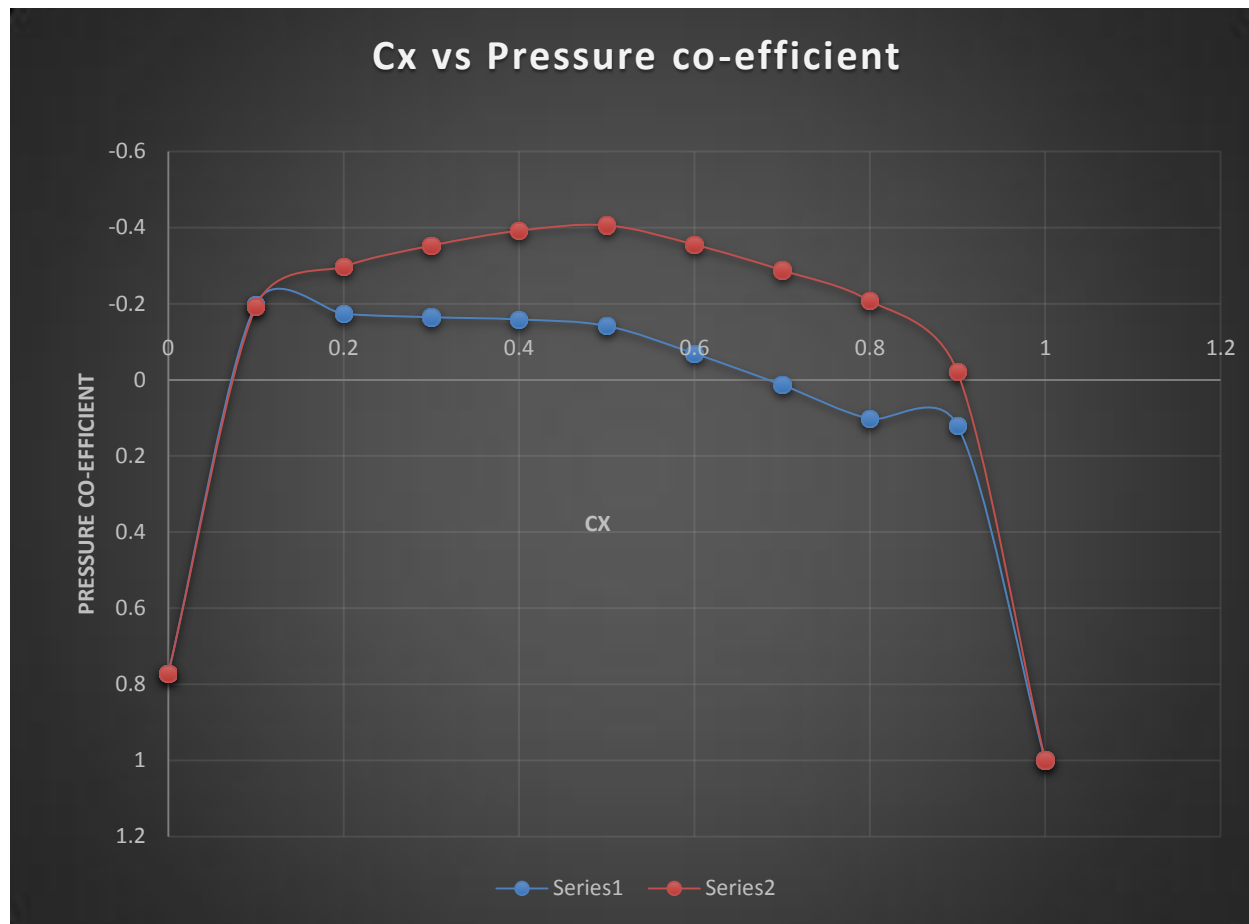
For $x=-.9$, $CL=.1061$, $\Delta v/V$ is multiplied by $1.82/6.79=0.268$. Lift developed by camber= $1*0.268=.268$

Lift to be developed= $0.1061-0.268=0.1619$. $\Delta v_a/V$ multiplied by 0.1619

for $x=-.9$ $CL=.1061$											
	Cx (From L.E)	v/V	$\Delta v_a/V$	$\Delta v/V$	v/V (Total) back	($\Delta p/q$) back	v/V (Total) face	($\Delta p/q$) face	($\Delta p/q$) face - ($\Delta p/q$) back	S.M	Product
1	0	0	-0.4836	0.074504	-0.4090913	0.83264431	0.409091	0.83264431	0	1	0
2	0.1	1.093	-0.07739	0.074504	1.0901158	-0.1883525	1.095884	-0.2009622	-0.01261	4	-0.05044
3	0.2	1.111	-0.05229	0.074504	1.1332103	-0.2841656	1.08879	-0.185463	0.098703	2	0.197405
4	0.3	1.121	-0.04031	0.074504	1.1551909	-0.334466	1.086809	-0.181154	0.153312	4	0.613248
5	0.4	1.128	-0.03206	0.074504	1.1704478	-0.3699481	1.085552	-0.1784236	0.191524	2	0.383049
6	0.5	1.127	-0.02607	0.074504	1.1754381	-0.3816547	1.078562	-0.1632958	0.218359	4	0.873436
7	0.6	1.099	-0.02056	0.074504	1.1529427	-0.3292769	1.045057	-0.0921448	0.237132	2	0.474264
8	0.7	1.064	-0.0157	0.074504	1.1227997	-0.2606792	1.0052	-0.0104276	0.250252	4	1.001006
9	0.8	1.023	-0.01149	0.074504	1.0860091	-0.1794158	0.959991	0.07841747	0.257833	2	0.515666
10	0.9	0.974	-0.00729	0.037252	1.0039665	-0.0079487	0.944034	0.10880075	0.116749	4	0.466998
11	1	0	0	0	0	1	0	1	0	1	0
Lift co-efficient, C_L =0.149154											4.474634

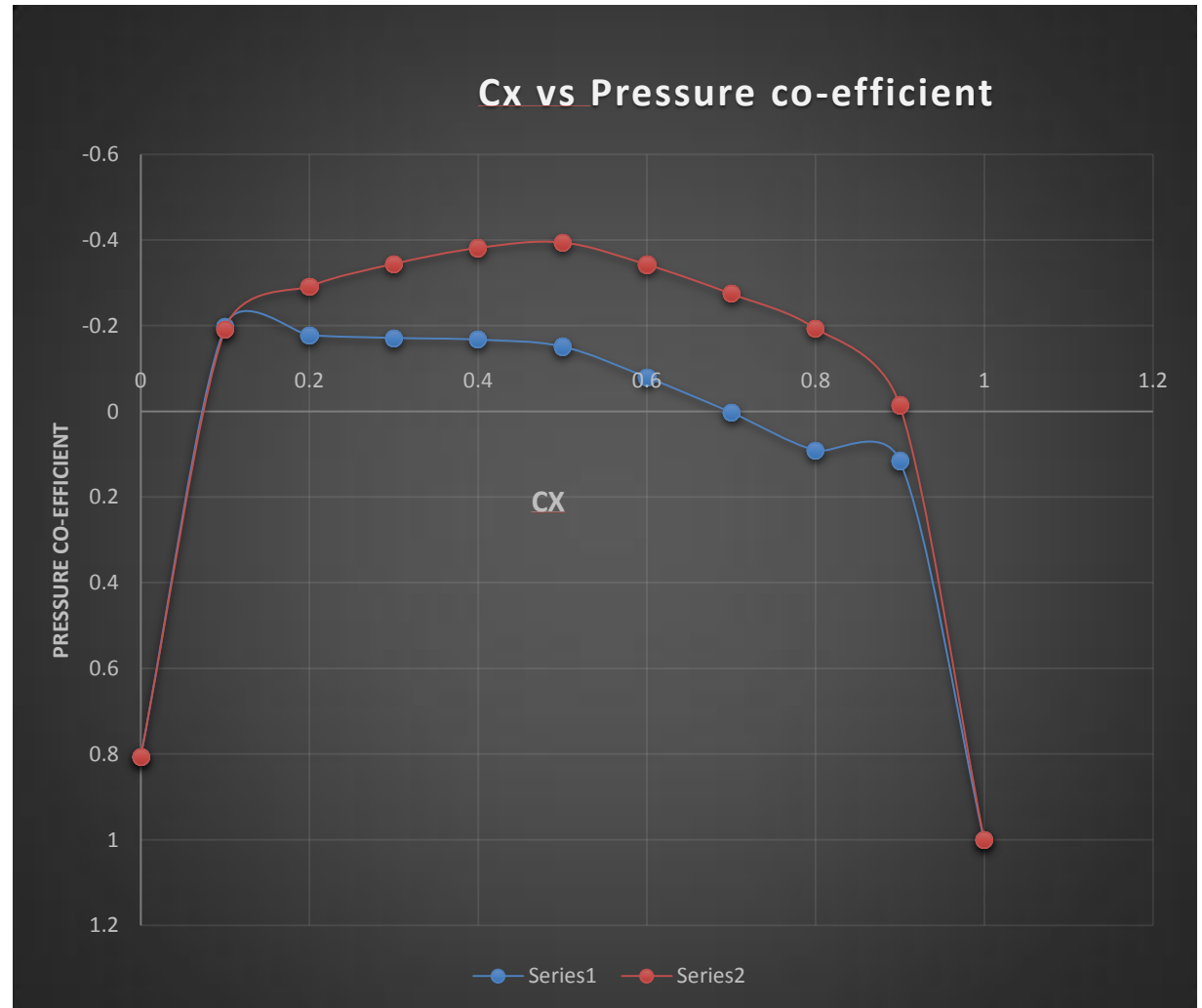
4.3.1 Graph of Cx vs Pressure co-efficient For 2.18% camber

	face	back
0	0.77152241	0.77152241
0.1	-0.1980051	-0.1912976
0.2	-0.173105	-0.2970939
0.3	-0.1643426	-0.3524594
0.4	-0.1585556	-0.3915453
0.5	-0.1413416	-0.4057994
0.6	-0.0689064	-0.355177
0.7	0.01358892	-0.2878112
0.8	0.1027271	-0.2072628
0.9	0.12029388	-0.0202485
1	1	1



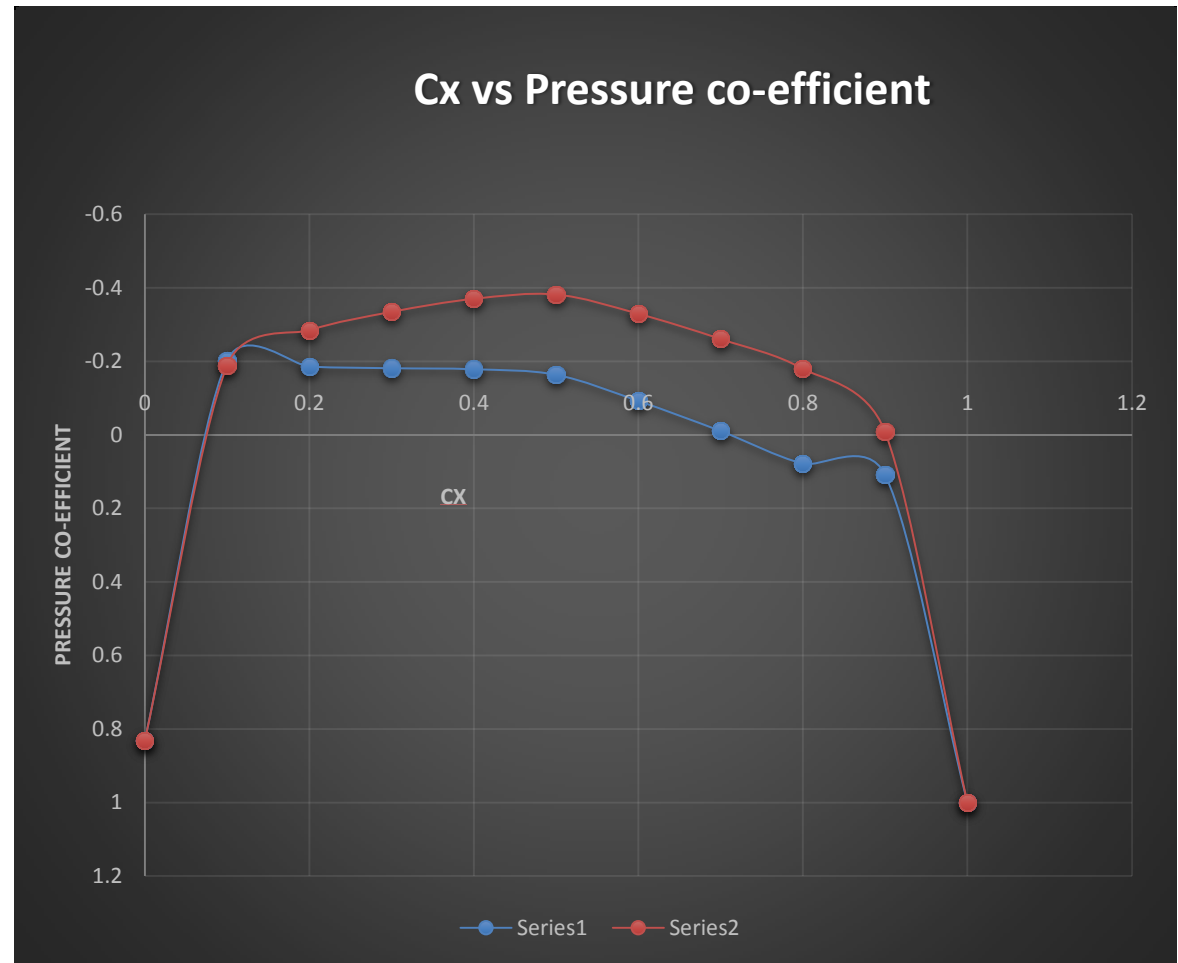
4.3.2 Graph of Cx vs Pressure co-efficient For 2% camber

	face	back
0	0.80682199	0.80682199
0.1	-0.1978182	-0.191484
0.2	-0.1780623	-0.2918918
0.3	-0.1717313	-0.3445205
0.4	-0.1676116	-0.3816573
0.5	-0.1515449	-0.3945234
0.6	-0.0798622	-0.3429004
0.7	0.00214645	-0.2748076
0.8	0.09105399	-0.193804
0.9	0.11479751	-0.0143471
1	1	1



4.3.3 Graph of Cx vs Pressure co-efficient For 1.82% camber

	face	back
0	0.83264431	0.83264431
0.1	-0.2009622	-0.1883525
0.2	-0.185463	-0.2841656
0.3	-0.181154	-0.334466
0.4	-0.1784236	-0.3699481
0.5	-0.1632958	-0.3816547
0.6	-0.0921448	-0.3292769
0.7	-0.0104276	-0.2606792
0.8	0.07841747	-0.1794158
0.9	0.10880075	-0.0079487
1	1	1



4.4 Summary of pressure calculation:

4.4.1 Combined Pressure co-efficient comparison for $x=.5$

	J=.833 (Full load) $\eta=.7058$ 2.18% Camber $x=.5$		J=.7 (Part load) $\eta=.6415$ 2.18% Camber $x=.5$		J=1 (Over load) $\eta=.6931$ 2.18% Camber $x=.5$	
	face	back	face	back	face	back
0	0.99168679	0.99168679	0.999671	0.999671	0.7715224	0.77152241
0.1	-0.0663313	-0.330255	-0.03051	-0.37092	-0.1980051	-0.1912976
0.2	-0.0842457	-0.3941205	-0.05977	-0.42218	-0.173105	-0.2970939
0.3	-0.0957936	-0.4284992	-0.0768	-0.45037	-0.1643426	-0.3524594
0.4	-0.104015	-0.452697	-0.08884	-0.47022	-0.1585556	-0.3915453
0.5	-0.0972278	-0.4556751	-0.08492	-0.46993	-0.1413416	-0.4057994
0.6	-0.0351695	-0.3937389	-0.02573	-0.40474	-0.0689064	-0.355177
0.7	0.03838285	-0.316479	0.045333	-0.32464	0.0135889	-0.2878112
0.8	0.12006145	-0.2275523	0.12493	-0.23332	0.1027271	-0.2072628
0.9	0.13119145	-0.0320548	0.134259	-0.0354	0.1202939	-0.0202485
1	1	1	1	1	1	1

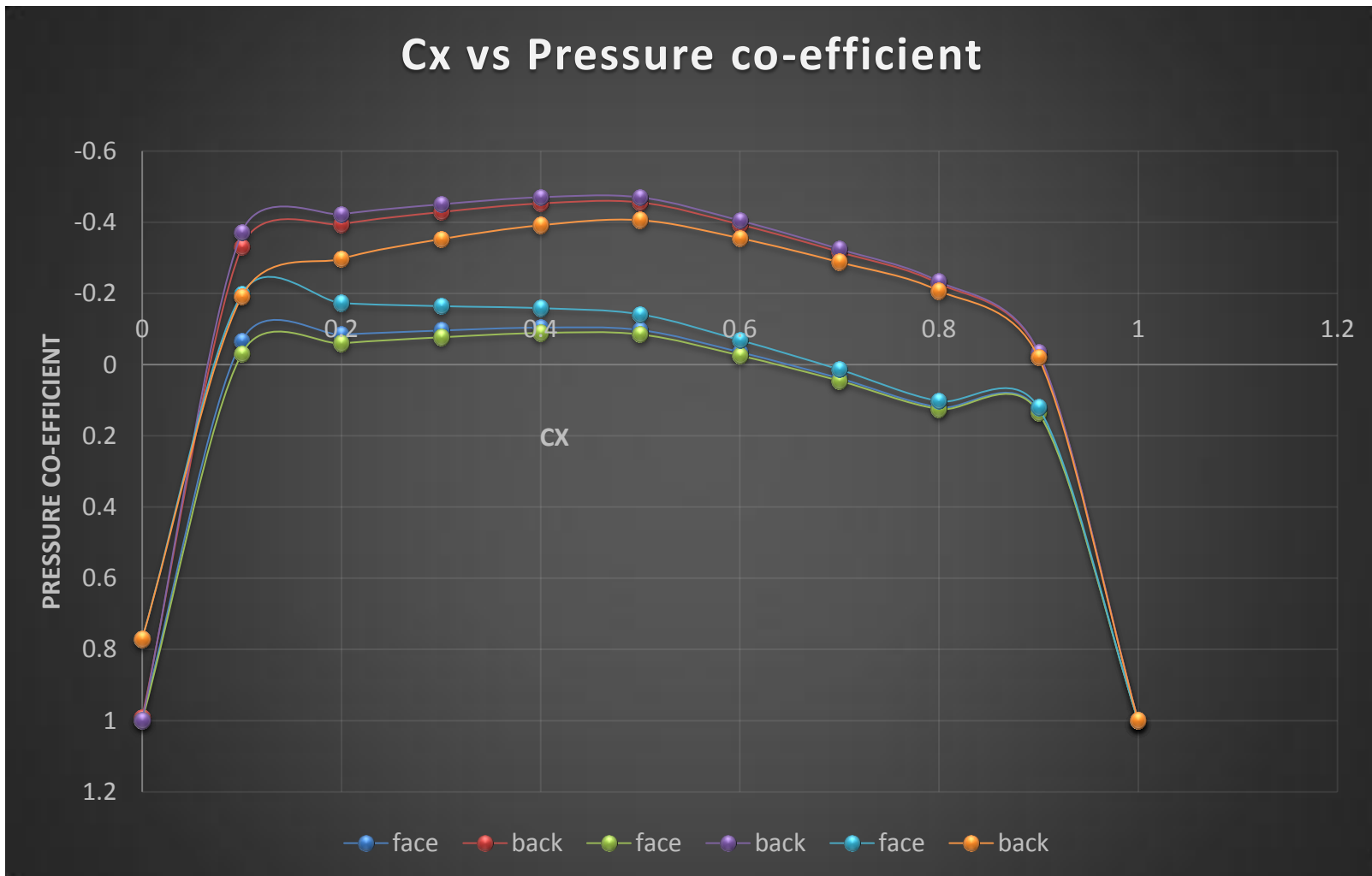


Figure 4.1: Cx vs Pressure co-efficient

4.4.2 Combined Pressure co-efficient comparison for $x=.7$

	J=.833 (Full load) $\eta=.7058$ 2% Camber $x=.7$		J=.7 (Part load) $\eta=.6415$ 2% Camber $x=.7$		J=1 (Over load) $\eta=.6931$ 2% Camber $x=.7$	
	face	back	face	back	face	back
0	0.94139752	0.94139752	0.994224	0.994224	0.806822	0.80682199
0.1	-0.1296565	-0.2614592	-0.07387	-0.32186	-0.1978182	-0.191484
0.2	-0.1321715	-0.3408818	-0.09428	-0.3828	-0.1780623	-0.2918918
0.3	-0.1363699	-0.3829607	-0.10705	-0.41571	-0.1717313	-0.3445205
0.4	-0.1394986	-0.4125964	-0.11612	-0.43889	-0.1676116	-0.3816573
0.5	-0.1288181	-0.4197711	-0.10988	-0.44118	-0.1515449	-0.3945234
0.6	-0.0624858	-0.362427	-0.04798	-0.37896	-0.0798622	-0.3429004
0.7	0.01491497	-0.2893273	0.025592	-0.3016	0.0021464	-0.2748076
0.8	0.09998064	-0.2040816	0.107455	-0.21276	0.091054	-0.193804
0.9	0.12038579	-0.0203475	0.125073	-0.02541	0.1147975	-0.0143471
1	1	1	1	1	1	1

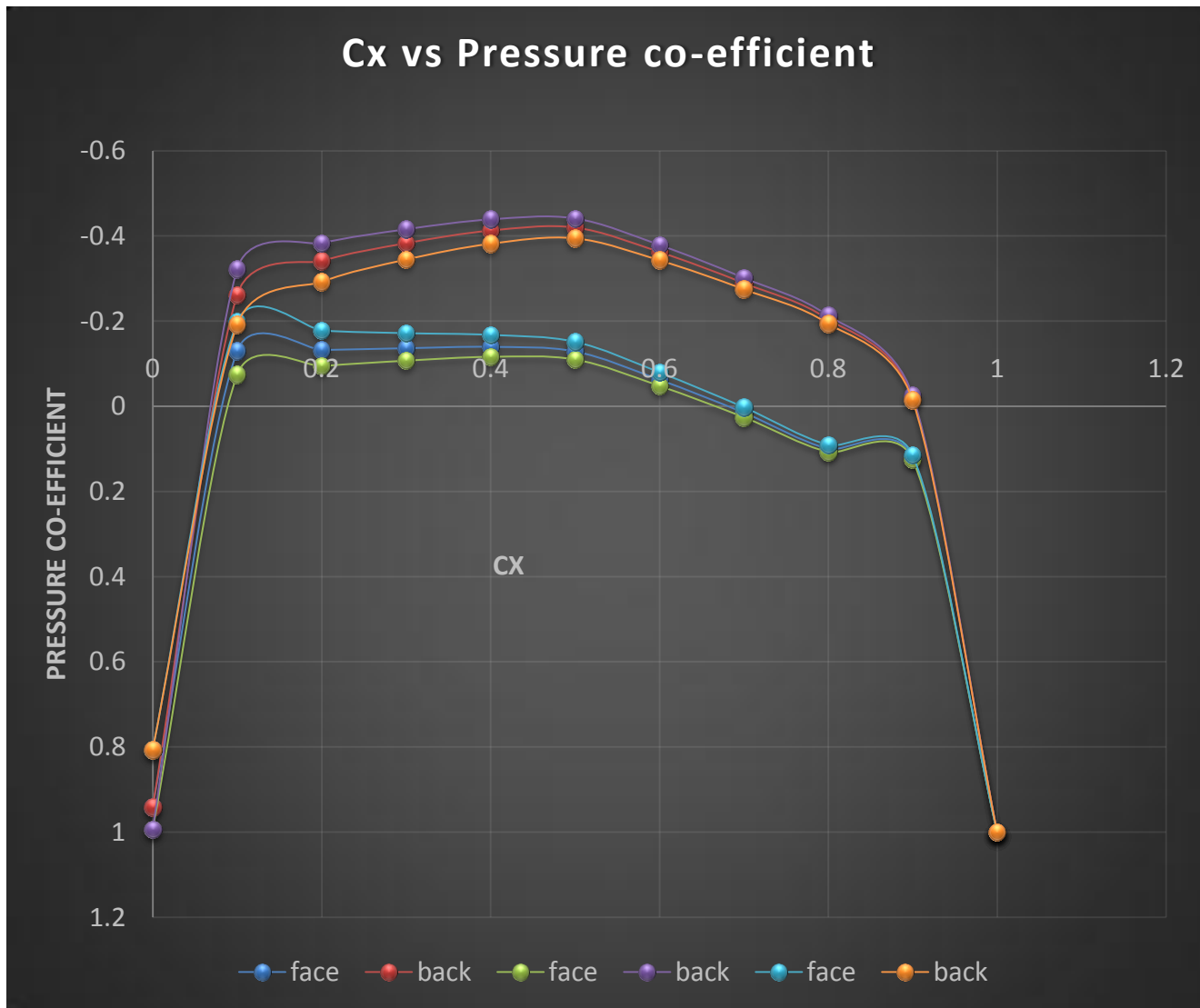


Figure 4.2: **Cx vs Pressure co-efficient**

4.4.3 Combined Pressure co-efficient comparison for x=.9

	J=.833 (Full load) η =.7058 1.82% Camber x=.9		J=.7 (Part load) η =.6415 1.82% Camber x=.9		J=1 (Over load) η =.6931 1.82% Camber x=.9	
	face	back	face	back	face	back
0	0.91971924	0.91971924	0.964298	0.964298	0.8326443	0.83264431
0.1	-0.1572604	-0.232632	-0.12499	-0.2664	-0.2009622	-0.1883525
0.2	-0.1560365	-0.31517	-0.13419	-0.33868	-0.185463	-0.2841656
0.3	-0.1584781	-0.3587954	-0.1416	-0.3772	-0.181154	-0.334466
0.4	-0.1603952	-0.3895308	-0.14695	-0.40432	-0.1784236	-0.3699481
0.5	-0.1487205	-0.3976351	-0.13784	-0.40969	-0.1632958	-0.3816547
0.6	-0.0809981	-0.3416343	-0.07267	-0.35095	-0.0921448	-0.3292769
0.7	-0.0022344	-0.2698662	0.003893	-0.27678	-0.0104276	-0.2606792
0.8	0.08414755	-0.1859171	0.088437	-0.19081	0.0784175	-0.1794158
0.9	0.1123741	-0.0117564	0.115052	-0.01462	0.1088008	-0.0079487
1	1	1	1	1	1	1

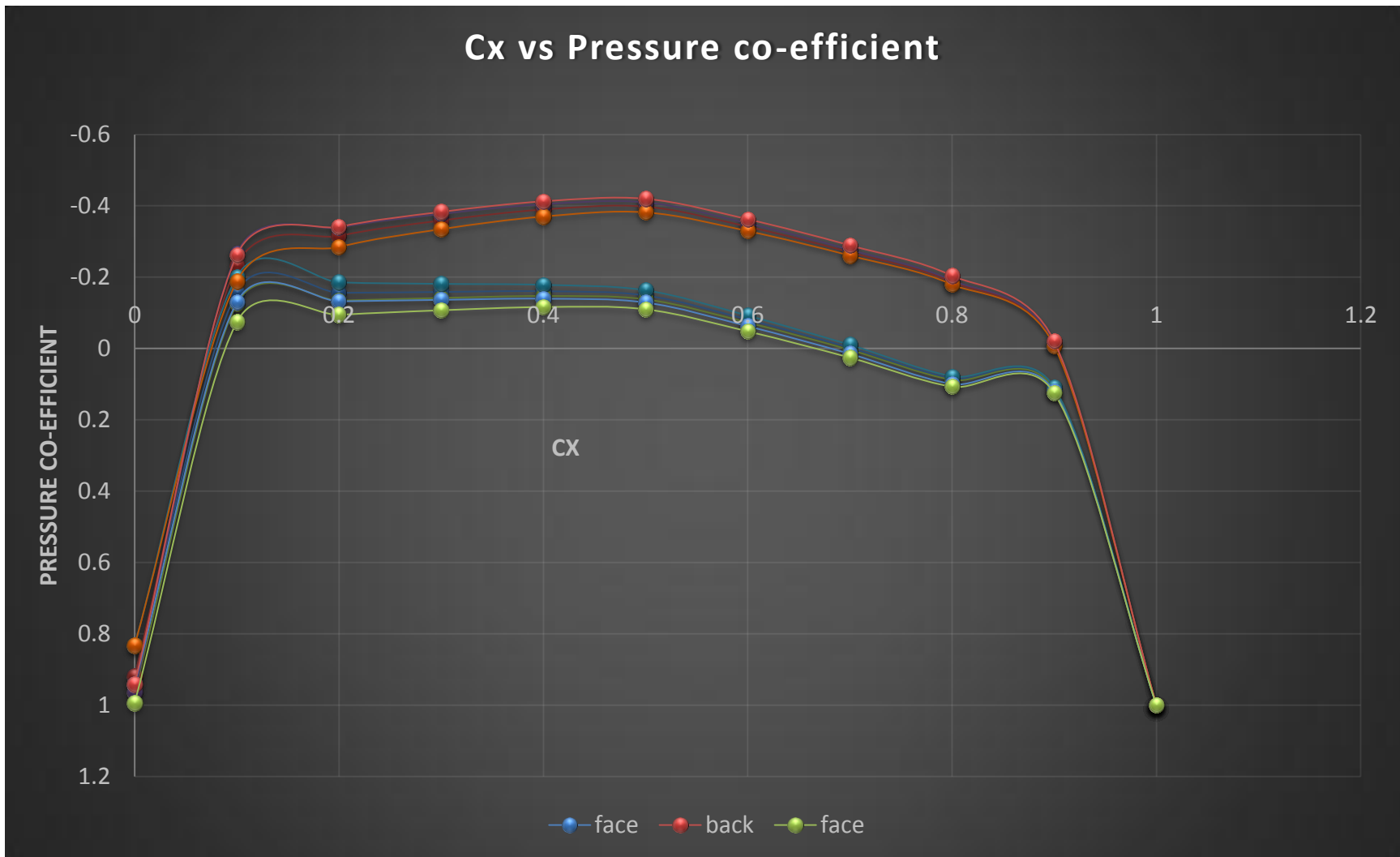


Figure 4.3: Cx vs Pressure co-efficient

CHAPTER 5

CAVITATION CHECK

A propeller produces the thrust by creating a difference between the pressures acting on the face and the back of the propeller blades, the pressures on the back of a blade section falling below the ambient pressure and the face rising above it. If the pressure at any point on the back of the blade falls to the vapor pressure, the water at that point begins to cavitate.

In chapter, the expression for pressure-coefficient at a point is defined as

$$\Delta p/q = 1 - [v/V(\text{total})]^2$$

The equation can be rewrite as

$$-\Delta p/q = 1 - [v/V(\text{total})]^2 - 1 = p_o - p_v / .5\rho v_R^2$$

Here pressure p_o is taken as the total static pressure(atmospheric plus hydrostatic pressure) at a point, p_v is the vapor pressure of the fluid and v_R as the relative velocity of water, then the condition for cavitation to occur may be written as,

$$-\Delta p/q = p_o - p_v / .5\rho v_R^2 = \sigma$$

Where σ is the local cavitation number.

The quantity $\Delta p/q$ is sometimes written as the pressure co-efficient C_p , so that the condition for cavitation to occur on a propeller blade section is that the minimum value of C_p be equal to the local cavitation number. This is normally written as follows:

$$-C_p(\text{minimum}) = \sigma$$

For a blade section at a non-dimensional radius $x=r/R$

$$-C_p(\text{minimum}) = (P_A + \rho g(h - xR) - P_v) / .5\rho v_R^2$$

Where,

P_A = Atmospheric Pressure= 101.325 KN/m^2

ρ = density of water= $1.025 \text{ KNs}^2/\text{m}^4$

g = Acceleration due to gravity= 9.81 m/s^2

h = Shaft immersion, m

x = Non-dimensional radius, ($=r/R$)

r = Section radius, m

R = Propeller radius, m

P_V =Vapor pressure of water= 1.704 KN/m^2

v_R = Relative velocity of water

$$= V_A \frac{\cos(\beta_i - \beta)}{\sin \beta}, m/s$$

A typical calculation for $-C_p$ (minimum)

$$J=0.7$$

$$x=0.5$$

$$\beta_i=24^0$$

$$v_R=2.4396 \text{ m/s}$$

$$P_A=101.325 \text{ KN/m}^2$$

$$\rho=1.025 \text{ KNs}^2/\text{m}^4$$

$$g=9.81 \text{ m/s}^2$$

$$h=.7D \text{ m}$$

$$x=0.5$$

$$R=0.3048/2=0.1524 \text{ m}$$

$$xR=0.5*0.1524=0.0762 \text{ m}$$

$$P_v=1.704 \text{ KN/m}^2$$

$$-C_p(\text{minimum})=\frac{[101.325+1.025*9.81(.021336-.0162-1.704)]}{.5*1.025*2.4396^2}$$

$$=33.11243$$

X	J	β(deg)	β_i(deg)	V_R(m/s)	Minimum Pressure (From graph)	-C_p ,min.	Remarks
0.5	0.7	24	31.122	2.4396	0.47022	33.11243	Cavitation doesn't occur
	0.833	27.93	33.07	2.1264	0.45568	43.58513	Cavitation doesn't occur
	1	32.488	32.548	1.862	0.4058	56.84194	Cavitation doesn't occur
0.7	0.7	17.646	23.248	3.315	0.44118	17.87892	Cavitation doesn't occur
	0.833	20.74	25.208	2.8153	0.41977	24.789	Cavitation doesn't occur
	1	24.45	24.503	2.42	0.39452	33.54885	Cavitation doesn't occur
0.9	0.7	13.897	19.014	4.147	0.40969	11.38981	Cavitation doesn't occur
	0.833	16.41	20.46	3.5309	0.39764	15.71136	Cavitation doesn't occur
	1	19.4775	21.927	2.675	0.38165	27.37392	Cavitation doesn't occur

CHAPTER 6

CONCLUSION

In this thesis, calculations on the basis of theory are done and how pressure is distributed is figured out. Finally several graphs were plotted for different blade sections at different radius from hub to the tip of a propeller blade.

By some basic analytical procedures we obtained the value of hydrodynamic pitch angle with induced velocity and with that we calculated thrust, torque and resulting propeller efficiency. The values obtained through calculations are validated by comparing it with that of Burrill analysis. This led to calculate pressure distribution which was validated by the lift co-efficient. These calculations are then followed by the procedure to figure out if cavitation is taking place. All the calculations are acceptable because the calculation is validated with result of Burrill. So this process can be used to calculate pressure distribution

NOMENCLATURE

Serial No	Symbol	Description
1	C	Propeller blade section chord length
2	C/D	Chord Ratio
3	C_D	Drag co-efficient
4	C_L	Lift co-efficient
5	$-C_p(\text{minimum})$	Minimum pressure co-efficient of a section
6	C_x	Non-dimensional chord wise position
7	D	Diameter of propeller, ($D=2R$)
8	dT	Elemental thrust force
9	dQ	Elemental torque force
10	EAR	Expanded area ratio
11	J	Advance co-efficient
12	K_T	Overall propeller thrust co-efficient
13	K_Q	Overall propeller torque co-efficient
14	k	Goldstein correction factor
15	dK_T/dx	Non-dimensional section thrust co-efficient
16	dK_Q/dx	Non-dimensional section torque co-efficient
17	m	Camber of a section
18	m/c	Camber ratio
19	n	Propeller Rotational speed
20	P	Pitch of propeller blade section
21	P/D	Pitch ratio
22	R	Propeller radius, ($R=D/2$)
23	t	Thickness of a section
24	t/D	Thickness ratio
25	V_A	Speed of advance
26	V_R	Resultant fluid velocity
27	x	Non-dimensional radius, ($=r/R$)
28	z	Number of blades
29	β	Advance angle
30	β_i	Hydrodynamic pitch angle
31	γ	Lift drag angle
32	ρ	Density of water
33	η	Propeller efficiency
34	$\Delta p/q$	Pressure co-efficient

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